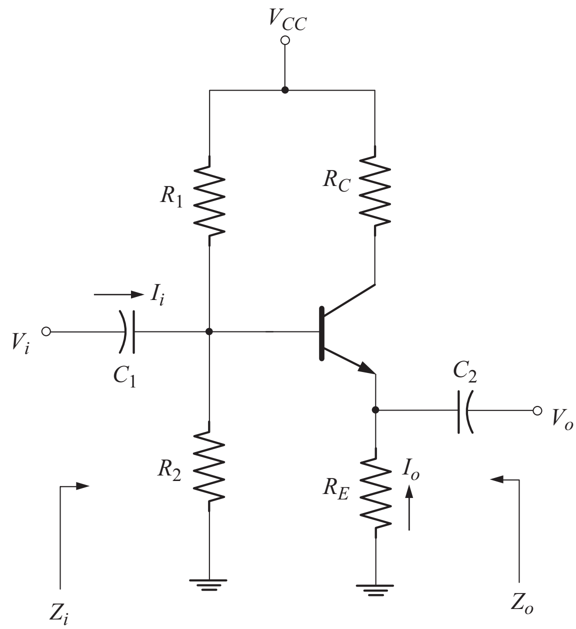


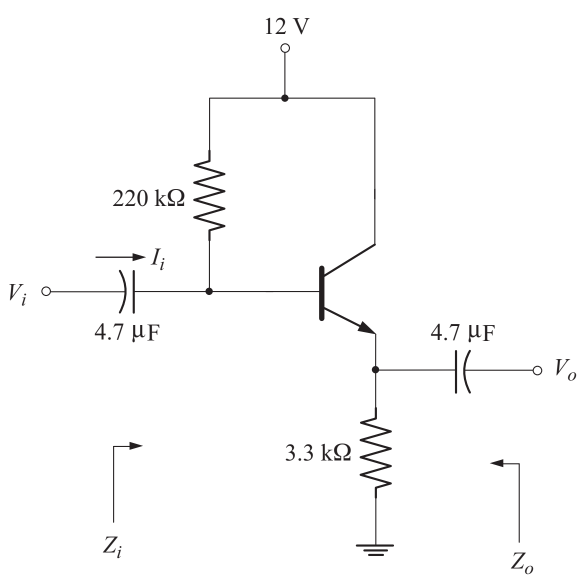
Fig. 3.60 Emitter follower using collector resistor

Example 3.11

For the emitter follower shown below calculate

(a) r_e (b) Z_i and Z_o (c) A_V and A_I

Take $\beta = 100$ and $r_o = \infty$.



Solution

(a)
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1+\beta)\, R_E}$$
$$= \frac{12\text{ V} - 0.7\text{ V}}{220\text{ k}\Omega + (101)(3.3\text{ k}\Omega)} = 20.42\ \mu\text{A}$$
$$I_E = (1+\beta)\, I_B = (101)\,(20.42\ \mu\text{A}) = 2.062\text{ mA}$$
$$r_e = \frac{26\text{mV}}{I_E} = \frac{26\text{mV}}{2.062\text{mA}} = 12.61\ \Omega$$

(b)
$$Z_b = \beta\, r_e + (1+\beta)\, R_E$$
$$= (100)\,(12.61\ \Omega) + (101)\,(3.3\text{ k}\Omega) = 334.56\text{ k}\Omega$$
$$Z_i = R_B \,\|\, Z_b = 220\text{ k}\Omega \,\|\, 334.56\text{ k}\Omega = 132.72\text{ k}\Omega$$
$$Z_o = R_E \,\|\, r_e = 3.3\text{ k}\Omega \,\|\, 12.61\ \Omega = 12.56\ \Omega$$

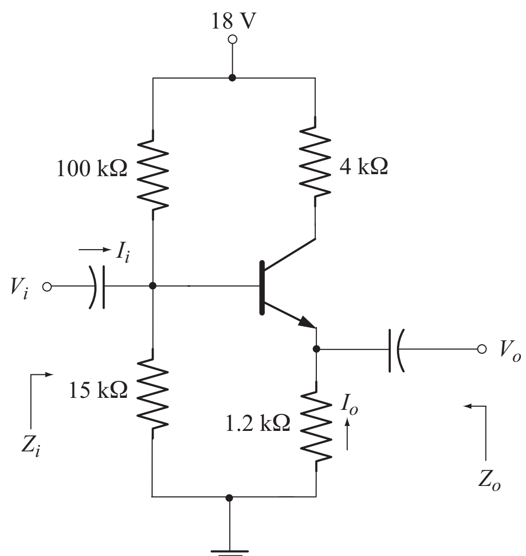
(c)
$$A_V = \frac{R_E}{R_E + r_e} = \frac{3.3\text{ k}\Omega}{3.3\text{ k}\Omega + 12.61\ \Omega} = 0.996$$
$$A_I = -\frac{A_V\, Z_i}{R_E} = -\frac{(0.996)\,(132.72\text{ k}\Omega)}{(3.3\text{ k}\Omega)} = -40$$

Example 3.12

For the circuit shown below calculate

(a) r_e (b) Z_i and Z_o (c) A_V and A_I

Take $\beta = 100$ and $r_o = 50\text{ k}\Omega$



Solution

(a) Using exact analysis for voltage divider bias circuit (since $\beta R_E < 10 R_2$) we get

$$I_E = 1.23 \text{ mA}$$

$$r_e = \frac{26\,\text{mV}}{I_E} = \frac{26\,\text{mV}}{1.23\,\text{mA}} = 21.13\ \Omega$$

(b)

$$Z_b = \beta(r_e + R_E) = 100\ (21.13\ \Omega + 1.2\ \text{k}\Omega) = 122.11\ \text{k}\Omega$$

$$Z_i = R_B \parallel Z_b$$

$$R_B = 100\ \text{k}\Omega \parallel 15\ \text{k}\Omega = 13.04\ \text{k}\Omega$$

$$Z_i = 13.04\ \text{k}\Omega \parallel 122.11\ \text{k}\Omega = 11.78\ \text{k}\Omega$$

$$Z_o = R_E \parallel r_e = 1.2\ \text{k}\Omega \parallel 21.13\ \Omega = 20.76\ \Omega$$

(c)

$$A_V = \frac{R_E}{r_e + R_E} = \frac{1.2\ \text{k}\Omega}{21.13\ \Omega + 1.2\ \text{k}\Omega} = 0.983$$

$$A_I = -\frac{A_V Z_i}{R_E} = -\frac{(0.983)(11.78\ \text{k}\Omega)}{(1.2\ \text{k}\Omega)} = -9.65$$

3.27 COMMON-BASE CONFIGURATION

Figure 3.61 shows the circuit of common-base configuration.

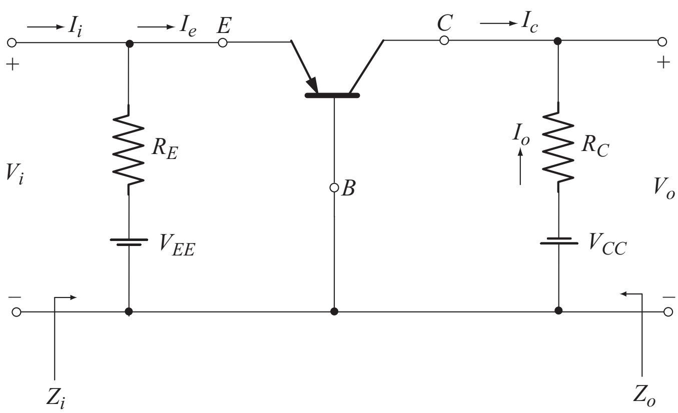


Fig. 3.61 Common-base configuration

The ac equivalent circuit is drawn in Fig. 3.62 by replacing the transistor with its r_e equivalent model. The output impedance r_o of the transistor in CB configuration is typically in the mega ohm range. Hence it is ignored in parallel with R_C.

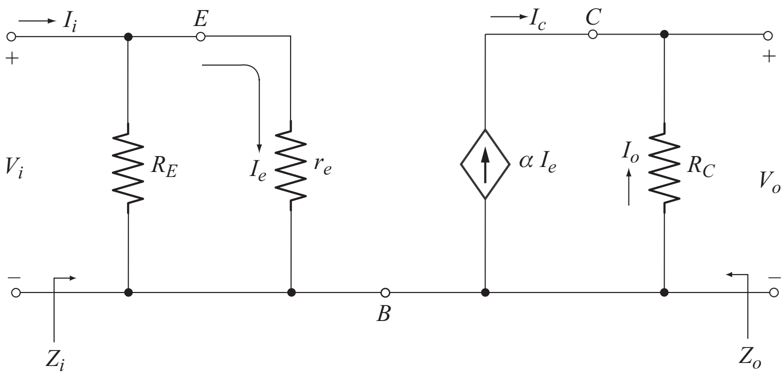


Fig. 3.62 *AC equivalent circuit with r_e model*

Input Impedance (Z_i)

From the ac equivalent circuit of Fig. 3.62 we find that Z_i is given by the parallel combination of R_E and r_e.

$$Z_i = \frac{V_i}{I_i} = R_E \parallel r_e \tag{3.95}$$

Since $r_e \ll R_E$

$$Z_i \approx r_e \tag{3.96}$$

Output Impedance (Z_o)

To find Z_o, we have to set $V_i = 0$. with $V_i = 0$, $I_e = 0$ and therefore $\alpha I_e = 0$. Thus the current source should be replaced by its open circuit equivalent as shown in Fig. 3.63.

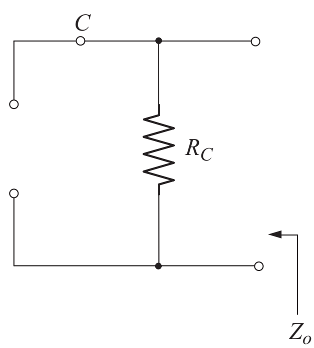


Fig. 3.63 Circuit to find Z_0

From the circuit of Fig. 3.63 we find that

$$Z_o = R_C \tag{3.97}$$

Voltage Gain (A_V)

Refer the circuit of Fig. 3.62

$$V_o = -I_o R_C$$
$$= -(-I_c) R_C \qquad (\because I_c = -I_o)$$
$$= I_c R_C$$
$$V_i = I_e r_e$$
$$= \frac{I_c}{\alpha} r_e \qquad (\because I_c = \alpha I_e)$$

Now $$A_V = \frac{V_o}{V_i}$$
$$= [I_c R_C] \div \left[\frac{I_c r_e}{\alpha}\right]$$
$$A_V = \frac{\alpha R_C}{r_e} \tag{3.98}$$
$$A_V \approx \frac{R_C}{r_e} \qquad \text{(Since } \alpha \approx 1) \tag{3.99}$$

Current Gain (A_I)

Since $r_e \ll R_E$, we can take $I_e \approx I_i$

$$I_o = -I_c = -\alpha I_e$$
$$A_I = \frac{I_0}{I_i} = -\frac{\alpha I_e}{I_e}$$
$$A_I = -\alpha \tag{3.100}$$

Since $\alpha \approx 1$, $$A_I \approx -1 \tag{3.101}$$

Phase Relationship

The positive sign for A_V reveals that, the input voltage V_i and the output voltage V_o are in phase.

Effect of r_o

For the CB configuration, the output resistance $r_o = 1/h_{ob}$ is typically in the megohm range and much larger than the parallel resistance R_C. Hence we can take

$$r_o \parallel R_C \approx R_C$$

3.28 IMPORTANT CHARACTERISTICS OF CB CONFIGURATION

Following are the important characteristics of CB configuration.

- Very low input impedance.

$$Z_i = R_E \parallel r_e \approx r_e$$

- High voltage gain.

$$A_V \approx \frac{R_C}{r_e} \gg 1 \qquad [\text{Since } r_e \ll R_C]$$

- Approximately unity current gain.
- High output impedance, since r_o is typically in the megohm range.
- Input and output voltages are in phase.

3.29 APPLICATIONS OF CB CONFIGURATION

Due to its low input impedance CB amplifier overloads most signal sources. Thus CB amplifier is not used at low frequencies. It is mainly used for high frequency applications (above 10 MHz) where low source impedances are common.

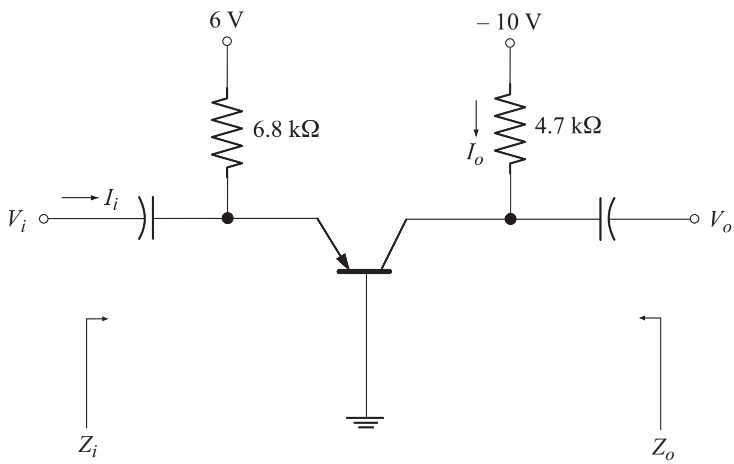
A common base-circuit is also used to couple a low impedance current source to a high impedance load. This application is called impedance matching.

Example 3.13

For the common–base configuration shown below calculate

(a) r_e (b) Z_i and Z_o (c) A_V and A_I

Take $\beta = 499$ and $r_o = 1\ \text{M}\Omega$.



Solution

$$\alpha = \frac{\beta}{1+\beta} = 0.998$$

(a)

$$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{6\text{ V} - 0.7\text{ V}}{6.8\text{ k}\Omega} = 0.779\text{ mA}$$

$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{0.779\text{ mA}} = 33.38\ \Omega$$

(b)
$$Z_i = R_E \parallel r_e = 6.8\text{ k}\Omega \parallel 33.38\ \Omega = 33.21\ \Omega$$
$$Z_o = R_C = 4.7\text{ k}\Omega$$

(c)
$$A_V = \frac{\alpha R_C}{r_e} = \frac{(0.998)(4.7\text{ k}\Omega)}{33.38\ \Omega} = 140.52$$
$$A_I = -\alpha = -0.998.$$

3.30 COLLECTOR FEEDBACK CONFIGURATION

Figure 3.64 shows the circuit of collector feedback configuration. Note that feedback is given from collector to base through R_F to increase the stability of the system.

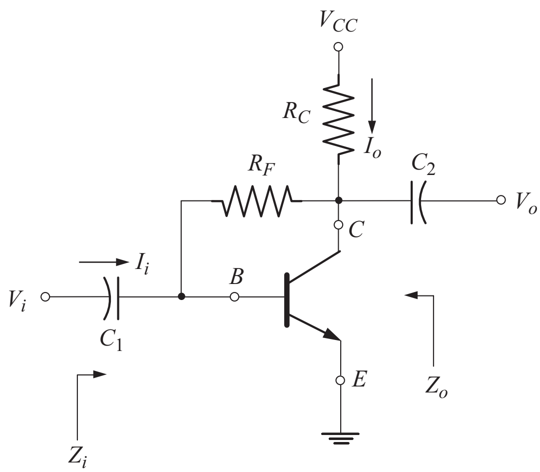


Fig. 3.64 Collector feedback configuration

The ac equivalent circuit is drawn by reducing V_{CC} to zero and replacing C_1 and C_2 by their short circuit equivalents as shown in Fig. 3.65. Note that the configuration is common-emitter since the emitter terminal is present in both input and output circuits.

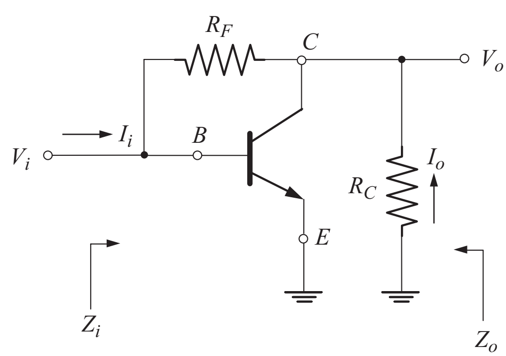


Fig. 3.65 AC equivalent circuit

Let us replace the transistor by its r_e equivalent model as shown in Fig. 3.66. Note that in the r_e model, the output resistance r_o of the transistor is not included, for simplicity.

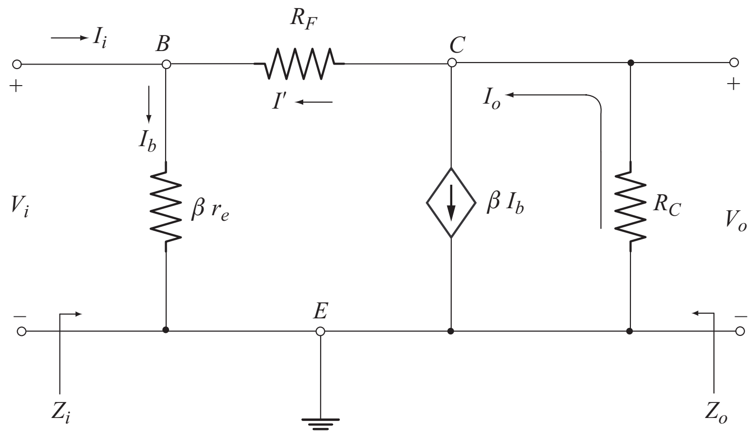


Fig. 3.66 *AC* Equivalent circuit with r_e model

Input Impedance (Z_i)

$$V_i = I_b \beta r_e \tag{3.102}$$

Applying KCL at the base node we have

$$I_b = I_i + I' \tag{3.103}$$

$$I' = \frac{V_0 - V_i}{R_F}$$

$$I' = \frac{V_0}{R_F} - \frac{V_i}{R_F} \tag{3.104}$$

$$V_o = -I_o R_C \tag{3.105}$$

Applying KCL at the collector node we have

$$I_o = \beta I_b + I'$$

Usually $\beta I_b \gg I'$

Thus $I_o \approx \beta I_b$

Using this relation in Equation (3.105) we have

$$V_o = -\beta I_b R_C$$

Substituting for I_b from Equation (3.102)

$$V_o = -\beta\left(\frac{V_i}{\beta r_e}\right) R_C = -V_i \frac{R_C}{r_e}$$

Substituting this relation in Equation (3.104) we have

$$I' = -V_i \frac{R_C}{r_e R_F} - \frac{V_i}{R_F}$$

$$I' = -\frac{1}{R_F}\left[1 + \frac{R_C}{r_e}\right] V_i \tag{3.106}$$

Let us substitute this relation in Equation (3.103)

$$I_b = I_i - \frac{1}{R_F}\left[1+\frac{R_C}{r_e}\right] V_i$$

$$\frac{V_i}{\beta r_e} = I_i - \frac{1}{R_F}\left[1+\frac{R_C}{r_e}\right] V_i$$

$$V_i = I_i \beta r_e - \frac{\beta r_e}{R_F}\left[1+\frac{R_C}{r_e}\right] V_i$$

$$V_i\left[1 + \frac{\beta r_e}{R_F}\left[1 + \frac{R_C}{r_e}\right]\right] = I_i \beta r_e$$

Now
$$Z_i = \frac{V_i}{I_i} = \frac{\beta r_e}{1 + \frac{\beta r_e}{R_F}\left[1 + \frac{R_C}{r_e}\right]} \tag{3.107}$$

Usually,
$$R_C \gg r_e \quad \Rightarrow \quad 1 + \frac{R_C}{r_e} \approx \frac{R_C}{r_e}$$

Hence
$$Z_i \approx \frac{\beta r_e}{1 + \frac{\beta R_C}{R_F}}$$

or
$$Z_i = \frac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_F}} \tag{3.108}$$

Output Impedance (Z_o)

To find Z_o, let us set $V_i = 0$. With $V_i = 0$, $I_b = 0$ and hence $\beta I_b = 0$. The circuit to find Z_o is shown in Fig. 3.67.

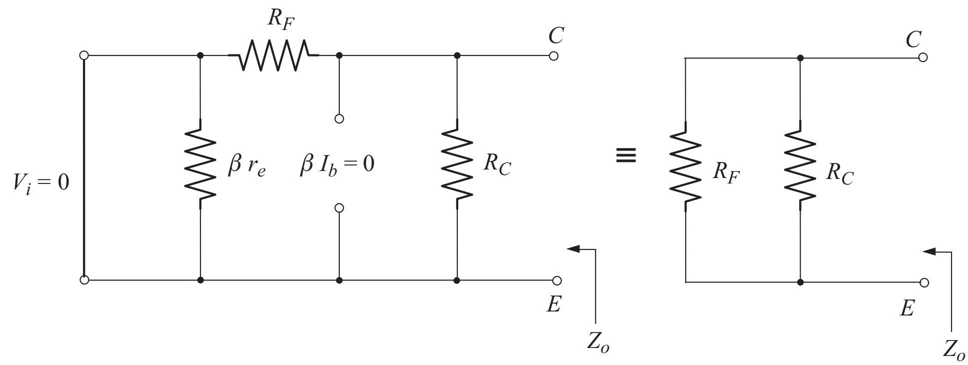


Fig. 3.67 Circuit to find Z_0

Z_o is given by the parallel combination of R_F and R_C

$$\therefore \qquad Z_o = R_F \parallel R_C \tag{3.109}$$

Voltage Gain **(*A_V*)**

Refer the circuit of Fig. 3.66

$$I_o = \beta I_b + I' \approx \beta I_b \qquad \text{Since } \beta I_b \gg I'$$

$$V_0 = -I_o R_C = -\beta I_b R_C$$

$$V_i = I_b \beta r_e \qquad \text{(From Equation (3.102))}$$

$$A_V = \frac{V_0}{V_i} = -\frac{\beta I_b R_C}{I_b \beta r_e}$$

$$A_V = -\frac{R_C}{r_e} \tag{3.110}$$

Current Gain **(*A_I*)**

$$V_o = -I_o R_C \quad \Rightarrow \quad I_o = \frac{-V_0}{R_C}$$

$$V_i = I_i Z_i \quad \Rightarrow \quad I_i = \frac{V_i}{Z_i}$$

$$\text{Now} \qquad A_I = \frac{I_o}{I_i} = \frac{-V_o/R_C}{V_i/Z_i}$$

$$= -\frac{V_0}{V_i}\frac{Z_i}{R_C}$$

$$\therefore \qquad A_I = -A_V \frac{Z_i}{R_C} \tag{3.111}$$

Phase Relationship

The negative sign for A_V in Equation (3.110) implies a 180° phase shift between V_o and V_i.

Note : If an emitter resistance R_E which is unbypassed and included in the circuit of Fig. 3.64 the expressions for Z_i and A_V gets modified as follows.

$$\therefore \qquad Z_i = \frac{R_E}{\frac{1}{\beta} + \frac{R_E + R_C}{R_F}}$$

$$Z_o = R_C \parallel R_F \qquad \text{(same as before)}$$

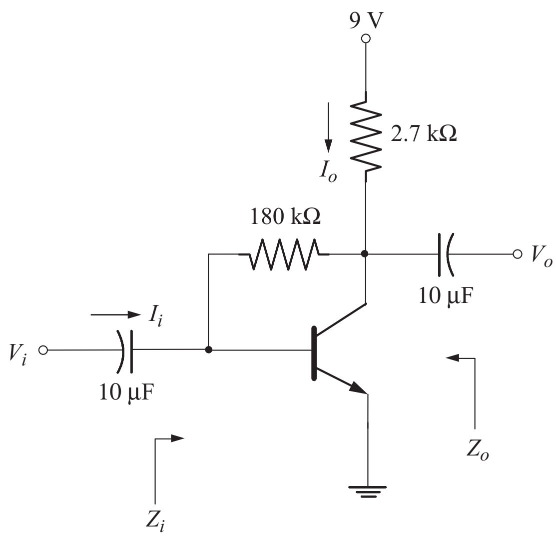
$$A_V = -\frac{R_C}{R_E}$$

Example 3.14

For the circuit shown below calculate

(a) r_e (b) Z_i and Z_o (c) A_V and A_I

Take $\beta = 200$ and $r_o = 60\ \text{k}\Omega$.



Solution

(a)
$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = \frac{9\ \text{V} - 0.7\ \text{V}}{180\ \text{k}\Omega + (200)(2.7\ \text{k}\Omega)}$$

$$= 11.53\ \mu\text{A}$$

$$I_E = (1 + \beta)\, I_B = (201)\,(11.53\ \mu\text{A}) = 2.32\ \text{mA}$$

$$r_e = \frac{26\,\text{mV}}{I_E} = \frac{26\,\text{mV}}{2.32\ \text{mA}} = 11.21\ \Omega$$

(b)
$$Z_i = \frac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_F}} = \frac{11.21\ \Omega}{\frac{1}{200} + \frac{2.7\ \text{k}\Omega}{180\ \text{k}\Omega}} = 560.5\ \Omega$$

$$Z_o = R_C \,||\, R_F = 2.7\ \text{k}\Omega \,||\, 180\ \text{k}\Omega = 2.66\ \text{k}\Omega$$

(c)
$$A_V = -\frac{R_C}{r_e} = -\frac{2.7\ \text{k}\Omega}{11.21\ \Omega} = -240.86.$$

$$A_I = -\frac{A_V\, Z_i}{R_C} = -\frac{(-240.86)\,(560.5\ \Omega)}{2.7\ \text{k}\Omega} = 50$$

◆ 3.31 COLLECTOR DC FEEDBACK CONFIGURATION

Figure 3.68 shows the circuit of collector dc feedback configuration. For dc operation capacitor C_3 acts as an open circuit. Hence the total dc feedback resistance is $R_{F1} + R_{F2}$. For ac operation C_3 acts as a short circuit. Thus R_{F1} appears on the input side and R_{F2} on the output side. The split feedback resistor arrangement improves the stability of the system.

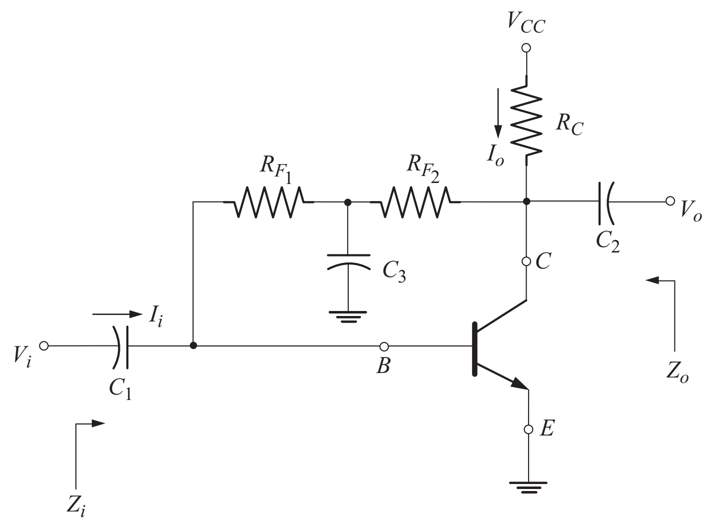


Fig. 3.68 Collector dc fedback configuration

The ac equivalent circuit is drawn by reducing V_{CC} to zero and replacing the capacitors by their short circuit equivalents as shown in Fig. 3.69.

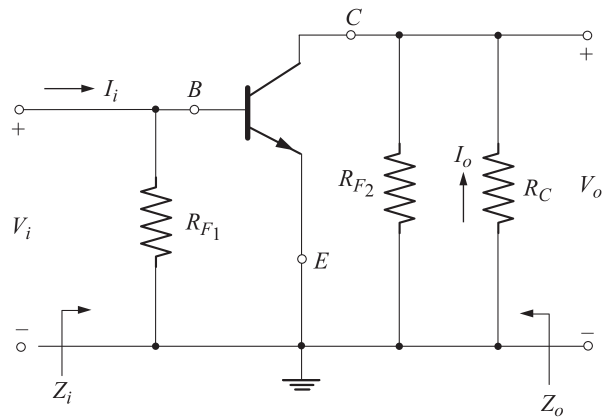


Fig. 3.69 *AC equivalent circuit*

Note that the configuration is common-emitter since the emitter terminal is present in both input and output circuits.

Let us replace the transistor by its r_e model as shown in Fig. 3.70.

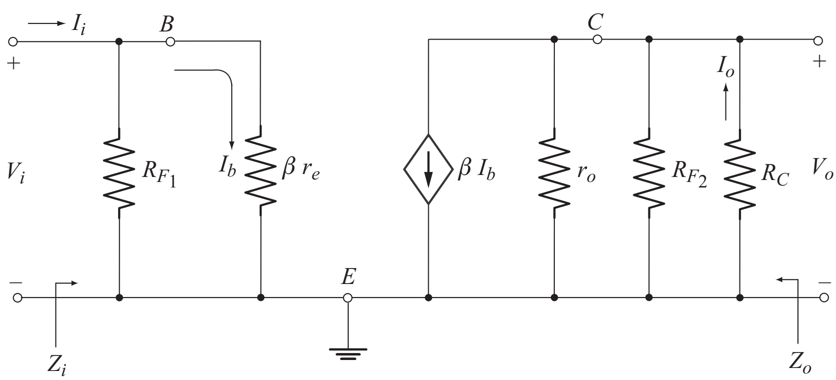


Fig. 3.70 *AC equivalent circuit with r_e model*

Input Impedance (Z_i)

As seen from the input circuit of Fig. 3.70, Z_i is given by the parallel combination of R_{F_1} and βr_e.

$$\therefore \qquad Z_i = \frac{V_i}{I_i} = R_{F_1} \| \beta r_e \tag{3.112}$$

Output Impedance (Z_o)

To find Z_o, let us set $V_i = 0$.

With $V_i = 0$, $I_i = 0$, $I_b = 0$ and therefore $\beta I_b = 0$.

The output equivalent circuit under this condition is shown in Fig. 3.71.

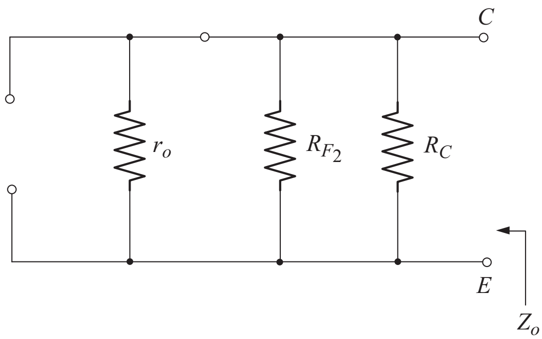


Fig. 3.71 Circuit to find Z_0

$$Z_o = r_o \parallel R_{F_2} \parallel R_C \tag{3.113}$$

If $r_o \geq 10\, R_C$,

$$r_o \parallel R_C \approx R_C$$

$$Z_o \approx R_{F_2} \parallel R_C \tag{3.114}$$

Voltage Gain **(A_V)**

Refer the circuit of Fig. 3.70.

Let us take,

$$R' = r_o \parallel R_{F_2} \parallel R_C \tag{3.115}$$

The simplified output equivalent circuit is shown in Fig. 3.72.

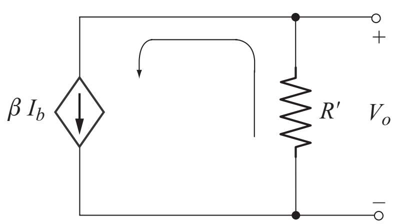


Fig. 3.72 Circuit to find V_o

$$V_o = -\beta I_b R'$$

$$V_i = \beta r_e I_b$$

$$A_V = \frac{V_0}{V_i} = -\frac{\beta I_b R'}{\beta r_e I_b}$$

$$A_V = -\frac{R'}{r_e} \tag{3.116}$$

For $r_o \geq 10\, R_C$,

$$R' = r_o \parallel R_{F_2} \parallel R_C$$

$$\approx R_{F_2} \parallel R_C$$

Now $$A_V \approx -\frac{R_{F_2} \| R_C}{r_e} \tag{3.117}$$

Current Gain (A_I)

$$V_o = -I_o R_C \quad \Rightarrow \quad I_o = -\frac{V_o}{R_C}$$

$$V_i = I_i Z_i \quad \Rightarrow \quad I_i = \frac{V_i}{Z_i}$$

Now $$A_I = \frac{I_0}{I_i} = \frac{-(V_0 / R_C)}{(V_i / Z_i)}$$

$$A_I = -A_V \frac{Z_i}{R_C} \tag{3.118}$$

Phase Relationship

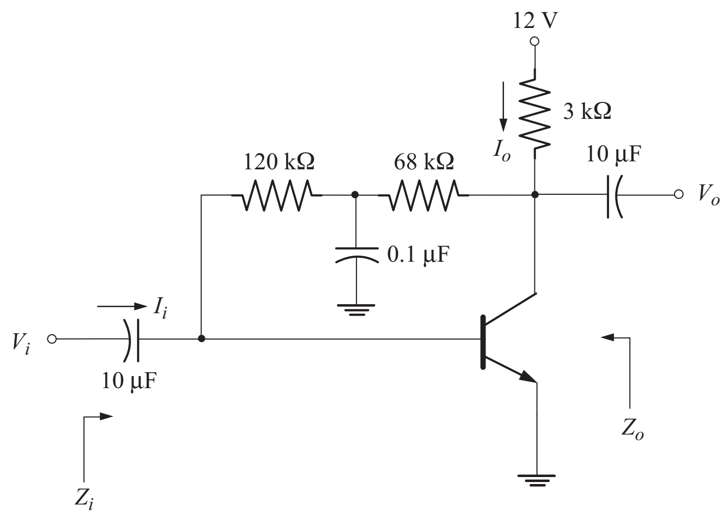
The negative sign for A_V clearly reveals that V_o and V_i are 180° out of phase.

Example 3.15

For the circuit shown below calculate

(a) r_e (b) Z_i and Z_o (c) A_V and A_I

Take $\beta = 140$ and $r_o = 30\ \text{k}\Omega$.



Solution

(a) For dc operation, 0.1 μF capacitor acts as an open circuit.
Thus the total dc feedback resistance is

$$R_F = 120\ \text{k}\Omega + 68\ \text{k}\Omega = 188\ \text{k}\Omega.$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = \frac{12\text{ V} - 0.7\text{ V}}{188\text{ k}\Omega + (140)(3\text{ k}\Omega)} = 18.59\ \mu\text{A}$$

$$I_E = (\beta + 1)\, I_B = (141)\,(18.59\ \mu\text{A}) = 2.62\text{ mA.}$$

$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{2.62\text{ mA}} = 9.92\ \Omega$$

(b)

$$Z_i = R_{F_1} \parallel \beta r_e = 120\text{ k}\Omega \parallel (140)\,(9.92\ \Omega) = 1.37\text{ k}\Omega$$

$$Z_o = r_o \parallel R_{F_2} \parallel R_C = 30\text{ k}\Omega \parallel 68\text{ k}\Omega \parallel 3\text{ k}\Omega = 2.62\text{ k}\Omega$$

(c)

$$A_V = -\frac{r_o \parallel R_{F_2} \parallel R_C}{r_e} = -\frac{2.62\text{ k}\Omega}{9.92\ \Omega} = -264.11$$

$$A_I = -\frac{A_V Z_i}{R_C} = -\frac{(-264.11)(1.37\text{ k}\Omega)}{3\text{ k}\Omega} = 120.61$$

3.32 EFFECTS OF R_S AND R_L

So for we have analyzed amplifiers with out the load resistance R_L and the source resistance R_S. Now let us proceed to analyze the amplifiers considering the effects of R_S and R_L. Now we will define three voltage gains for distinction as given below.

***No Load Voltage Gain* [*Unloaded voltage gain*]**

The no load voltage gain is defined with R_L not connected between the output terminals and with out considering R_S as shown in Fig. 3.73.

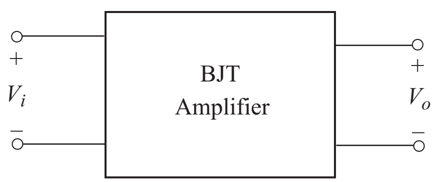


Fig. 3.73 Defining no load voltage gain

The no load voltage gain is given by

$$A_{V_{NL}} = \frac{V_o}{V_i} \tag{3.119}$$

It is important to note that $A_{V_{NL}}$ is exactly identical to A_V determined in the preceding sections. Thus in the subsequent analysis we shall use for $A_{V_{NL}}$, the relations derived earlier for A_V.

Loaded Voltage Gain

The loaded voltage gain is defined with R_L connected between the output terminals and without considering R_S as shown in Fig. 3.74.

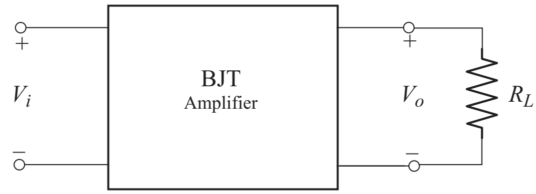


Fig. 3.74 Defining loaded voltage gain

The loaded voltage gain is given by

$$A_V = \frac{V_o}{V_i} \tag{3.120}$$

Though Equations (3.119) and (3.120) seem identical, the difference is that in the latter V_o is measured in the presence of R_L with same V_i.

Loaded Voltage Gain with Source Resistance

The loaded Voltage gain taking R_S into consideration is given by

$$A_{V_S} = \frac{V_o}{V_s} \tag{3.121}$$

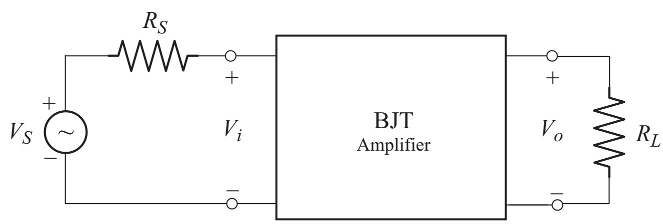


Fig. 3.75 Defining loaded voltage gain with R_S

3.32.1 Methods of Analysis of Amplifiers with R_S and R_L

There are two methods to analyze the amplifier networks with load and/or source resistance.

In the first method the r_e model is used as has been used in the previous sections.

In the second method a two port system approach is used.

First we will consider the r_e model approach.

3.33 FIXED BIAS COMMON-EMITTER AMPLIFIER WITH R_S AND R_L

Figure 3.76 shows the fixed bias CE amplifier with R_S and R_L. The ac equivalent circuit is drawn by replacing the transistor with its r_e model as shown in Fig. 3.77.

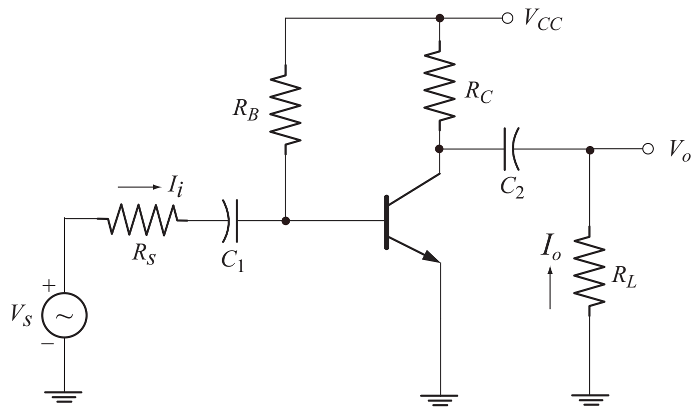


Fig. 3.76 Fixed bias CE configuration with R_S and R_L

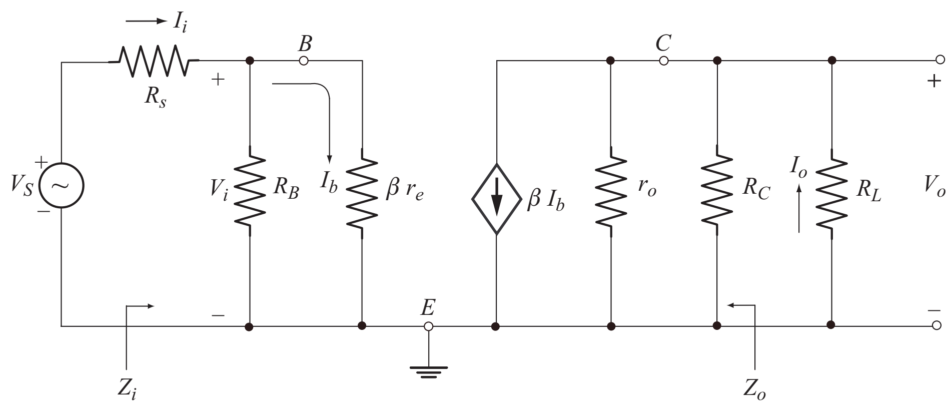


Fig. 3.77 AC equivalent circuit

Input Impedance (Z_i)

As seen from the input circuit of Fig. 3.77, the input impedance Z_i, is given by the parallel combination of R_B and $\beta\, r_e$.

$$\therefore \qquad Z_i = \frac{V_i}{I_b} = R_B \,\|\, \beta\, r_e \tag{3.122}$$

which is same as before.

Output Impedance (Z_o)

From the output circuit of Fig. 3.77 we find that, the output impedance is given by the parallel combination of r_o and R_C.

$$Z_o = r_o \,\|\, R_C \tag{3.123}$$

as before.

It is important to note that, Z_o does not take R_L into account.

Voltage Gain

Let R_L' represent the parallel combination of r_o, R_C and R_L.

Now $$R_L' = r_o \parallel R_C \parallel R_L \tag{3.124}$$

To simplify the analysis let us ignore the effect of r_o.

$$R_L' = R_C \parallel R_L$$

The simplified output equivalent circuit is shown in Fig. 3.78.

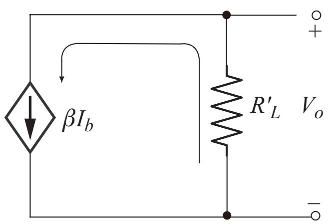


Fig. 3.78 Simplified output equivalent circuit

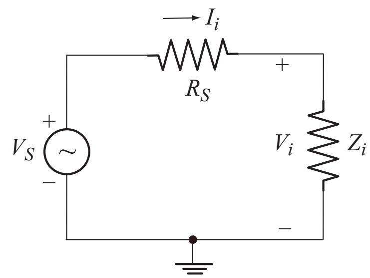


Fig. 3.79 Simplified input equivalent circuit

$$V_o = -\beta I_b R_L' \tag{3.125}$$

From the input circuit of Fig. 3.77

$$V_i = I_b \beta r_e \tag{3.126}$$

$$A_V = \frac{V_o}{V_i}$$

$$= -\frac{\beta I_b R_L'}{I_b \beta r_e}$$

$$A_V = -\frac{R_L'}{r_e} = -\frac{R_C \parallel R_L}{r_e} \tag{3.127}$$

To find A_{V_S} let us consider the input equivalent circuit shown in Fig. 3.79.

Using voltage division rule we have

$$V_i = V_S \frac{Z_i}{R_S + Z_i}$$

or $$\frac{V_i}{V_S} = \frac{Z_i}{R_S + Z_i} \tag{3.128}$$

$$A_{V_S} = \frac{V_o}{V_S} = \frac{V_o}{V_i}\frac{V_i}{V_S}$$

$$A_{V_S} = A_V \frac{Z_i}{Z_i + R_S} \tag{3.129}$$

The following interesting observations can be made with reference to the voltage gain.

(a)
$$A_V = -\frac{R_C \| R_L}{r_e}$$

For an unloaded amplifier, $R_L = \infty\ \Omega$

$$\therefore \quad R_C \| R_L = R_C$$

Thus
$$A_{V_{NL}} = -\frac{R_C}{r_e}$$

Since $R_C \| R_L < R_C$, it follows that

$$|A_V| < |A_{V_{NL}}|$$

The loaded voltage gain of an amplifier is always less than the no load voltage gain.

(b)
$$A_{V_S} = A_V \frac{Z_i}{R_S + Z_i}$$

Since $\dfrac{Z_i}{R_S + Z_i} < 1$, it follows that

$$A_{V_S} < A_V$$

The voltage gain obtained with R_S in place will always be less than that obtained under loaded or unloaded conditions.

Thus for the same configuration

$$|A_{V_{NL}}| > |A_V| > |A_{V_S}|$$

In total, the highest gain is obtained under no load condition and the lowest gain with a source impedance and load in place

(c) R_L' increases with increase in R_L. Therefore:

For a particular design, voltage gain increases with increase in R_L.

(d) With smaller R_S, $\dfrac{Z_i}{R_S + Z_i}$ approaches unity so that

A_{V_S} approaches A_V. Thus:

For a particular amplifier, the smaller the internal resistance of the signal source, the greater is the voltage gain.

Current Gain (A_I)

$$A_I = \frac{I_o}{I_i}$$

$$V_o = -I_o R_L \quad \Rightarrow \quad I_o = -\frac{V_o}{R_L}$$

$$V_i = I_i Z_i \Rightarrow I_i = \frac{V_i}{Z_i}$$

Now
$$A_I = \frac{-(V_o / R_L)}{(V_i / Z_i)}$$
$$= -\left(\frac{V_o}{V_i}\right)\left(\frac{Z_i}{R_L}\right)$$
$$A_I = -A_V\left(\frac{Z_i}{R_L}\right) \tag{3.130}$$

also
$$A_I = -\left.\frac{A_V Z_i}{R_C}\right|_{R_L = \infty}$$

Example 3.16

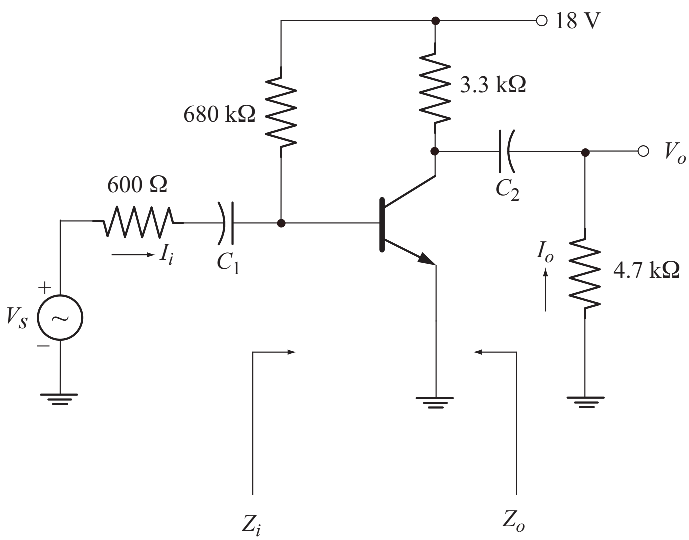
For the fixed bias configuration shown below

(a) Calculate $A_{V_{NL}}$, Z_i and Z_o

(b) Calculate A_V, A_{V_S} and A_I

(c) Calculate V_o if $V_S = 20$ mV

Take $\beta = 100$.



Solution

(a)
$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{18\,\text{V} - 0.7\,\text{V}}{680\,\text{k}\Omega} = 25.44\ \mu\text{A}$$
$$I_E = (1+\beta)\, I_B = (101)\,(25.44\ \mu\text{A}) = 2.57\ \text{mA}$$

$$r_e = \frac{26\,\text{mV}}{I_E} = \frac{26\,\text{mV}}{2.57\,\text{mA}} = 10.116\ \Omega$$

$$A_{V_{NL}} = -\frac{R_C}{r_e} = -\frac{3.3\,\text{k}\Omega}{10.116\,\Omega} = -326.22$$

$$Z_i = R_B \,\|\, \beta\, r_e = 680\ \text{k}\Omega \,\|\, (100)\,(10.116\ \Omega) = 1.01\ \text{k}\Omega$$

$$Z_o = R_C = 3.3\ \text{k}\Omega$$

(b)

$$A_V = -\frac{R_C \| R_L}{r_e} = -\frac{3.3\ \text{k}\Omega \,\|\, 4.7\ \text{k}\Omega}{10.116\ \Omega} = -191.65$$

$$A_{V_S} = A_V \frac{Z_i}{R_S + Z_i} = (-191.65)\left[\frac{1.01\,\text{k}\Omega}{0.6\,\text{k}\Omega + 1.01\,\text{k}\Omega}\right]$$

$$= -120.22$$

$$A_I = -A_V \frac{Z_i}{R_L} = -(-191.65)\,\frac{1.01\ \text{k}\Omega}{4.7\ \text{k}\Omega} = 41.18$$

(c)

$$A_{V_S} = \frac{V_o}{V_s} \quad \Rightarrow \quad V_o = A_{V_S} V_S$$

$$V_o = (-120.22)\,(20\ \text{mV}) = -2.\,4\ \text{V}$$

3.34 COMMON-EMITTER VOLTAGE DIVIDER BIAS CONFIGURATION, WITH R_S AND R_L

Figure 3.80 shows CE voltage divider bias configuration with R_S and R_L. Its ac equivalent circuit using r_e model is shown in Fig. 3.81.

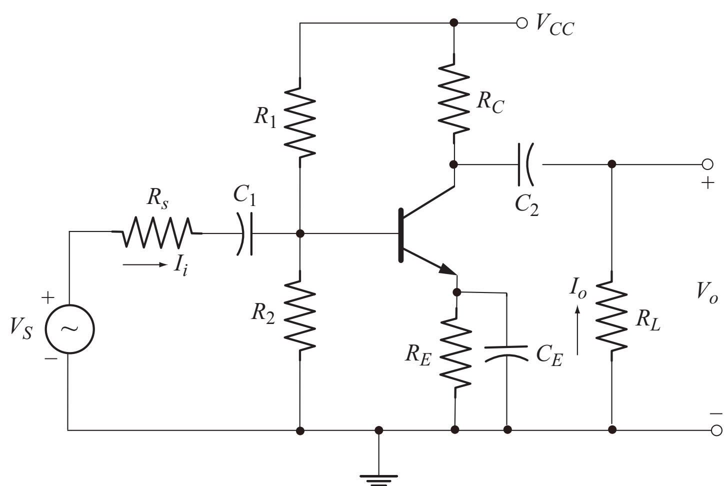


Fig. 3.80 CE voltage divider bias configuration with R_S and R_L

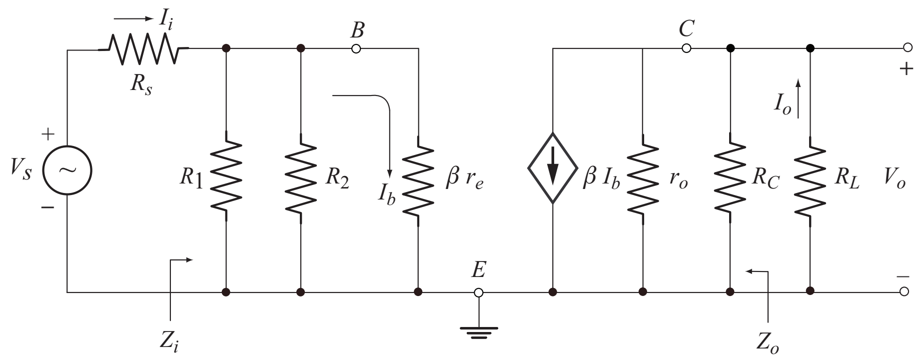


Fig. 3.81 Ac equivalent circuit using r_e model

Following the same procedure given in the previous section we obtain the following results.

- $$Z_i = R' \,||\, \beta r_e$$

  where $$R' = R_1 \,||\, R_2$$

- $$Z_o = r_o \,||\, R_C$$

- $$A_V = -\frac{R_C \,||\, R_L}{r_e}$$

- $$A_{V_{NL}} = -\frac{R_C}{r_e}$$

- $$A_{V_S} = \frac{V_o}{V_S} = A_V \frac{Z_i}{R_S + Z_i}$$

- $$A_I = \frac{I_o}{I_i} = -\frac{A_V Z_i}{R_L}$$

- $$A_I = -\left.\frac{A_V Z_i}{R_C}\right|_{R_L = \infty}$$

3.35 EMITTER-FOLLOWER CONFIGURATION WITH R_S AND R_L

Figure 3.82 shows the emitter-follower configuration with R_S and R_L. Its ac equivalent circuit using r_e model is shown in Fig. 3.83.

To simplify the analysis r_o is not included in the model.

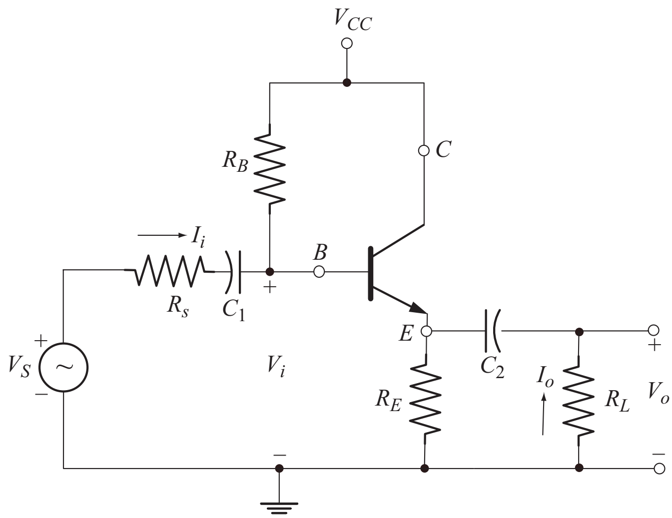


Fig. 3.82 Emitter-follower configuration with R_S and R_L

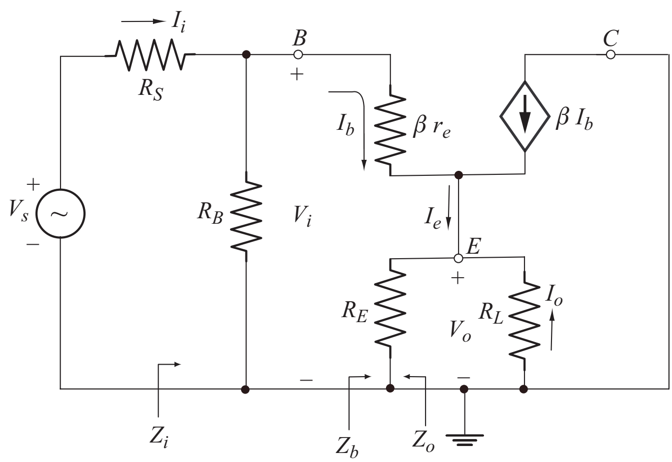


Fig. 3.83 Ac equivalent circuit using r_e model

The analysis given in section 3.24 can be readily applied to this circuit by replacing R_E with $R_E \parallel R_L$ in the equations of Z_b and A_V. The results are given below.

- $$Z_b = \beta r_e + (1+\beta)(R_E \parallel R_L) \approx \beta (R_E \parallel R_L) \tag{3.131}$$
- $$Z_i = R_B \parallel Z_b \tag{3.132}$$
- $$Z_o = r_e \parallel R_E \approx r_e \tag{3.133}$$
- $$A_V = \frac{R_E \parallel R_L}{r_e + R_E \parallel R_L} \tag{3.134}$$

- $$A_{V_{NL}} = \frac{R_E}{r_e + R_E} \tag{3.135}$$

- $$A_{V_S} = A_V \frac{Z_i}{Z_i + R_S} \tag{3.136}$$

- $$A_I = -\frac{A_V Z_i}{R_L} \tag{3.137}$$

- $$A_I = -\left.\frac{A_V Z_i}{R_C}\right|_{R_L = \infty}$$

◆ 3.36 CE EMITTER-BIAS CONFIGURATION WITH R_S AND R_L

Figure 3.84 shows CE emitter-bias configuration with R_S and R_L. For this circuit the analysis given in section 3.22 can be used by replacing R_C by $R_C \| R_L$ in the equation of A_V.

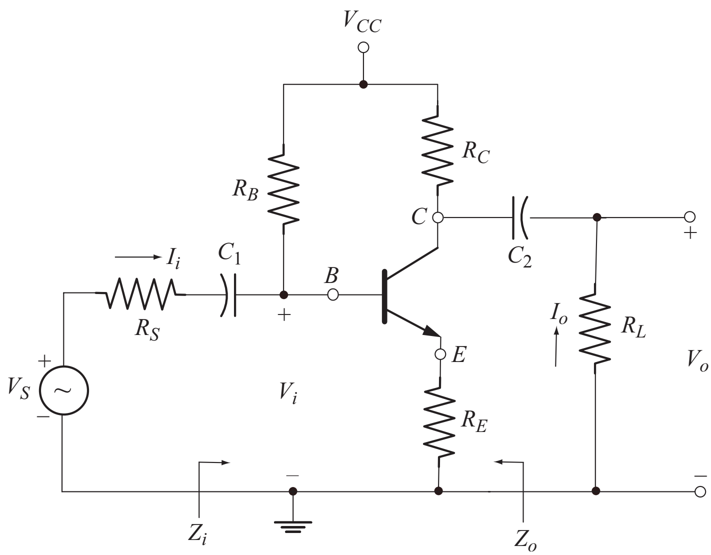


Fig. 3.84 CE emitter-bias configuration with R_S and R_L

The results are as follows:

- $$Z_b = \beta r_e + (1 + \beta) R_E \tag{3.138}$$
- $$Z_i = R_B \| Z_b \tag{3.139}$$
- $$Z_o = R_C \tag{3.140}$$
- $$A_V = -\frac{R_C \| R_L}{r_e + R_E} \tag{3.141}$$

- $$A_I = -\frac{A_V Z_i}{R_L} \tag{3.142}$$

- $$A_{V_{NL}} = -\frac{R_C}{r_e + R_E} \tag{3.143}$$

- $$A_I = -\left.\frac{A_V Z_i}{R_C}\right|_{R_L = \infty} \tag{3.144}$$

3.37 TWO-PORT SYSTEMS APPROACH

In two port system approach the amplifier is represented by a two port network as shown in Fig. 3.85. The parameters of the two port network are the input impedance Z_i, output impedance Z_o and the voltage gain $A_{V_{NL}}$. These parameters are specified for the unloaded condition. Using these results, the gain and impedances can be calculated under loaded conditions.

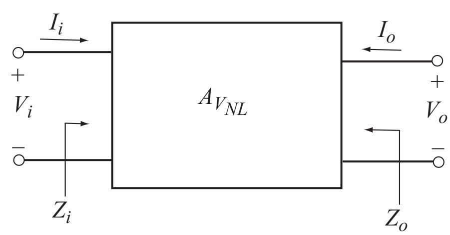


Fig. 3.85 Two - port system

The input Voltage V_i and the input current I_i are related by $V_i = I_i Z_i$. Therefore, between the input terminals, the amplifier can be represented by an impedance Z_i.

Under no load condition

$$A_{V_{NL}} = \frac{V_o}{V_i} \quad \Rightarrow \quad V_o = A_{V_{NL}} V_i$$

Thus at the output side, the amplifier can be represented by a controlled voltage source, $A_{V_{NL}} V_i$. Also it is appropriate to place the output impedance Z_o in series with the controlled voltage source.

The voltage source $A_{V_{NL}} V_i$ in series with the impedance Z_o is referred to as Thevenin equivalent circuit. We oftenly write

$$V_{Th} = A_{V_{NL}} V_i$$

and

$$Z_{Th} = Z_o$$

where V_{Th} is the open circuit voltage or the Thevenin voltage or no load output voltage and Z_{Th} is the Thevenins impedance.

For BJT and FET amplifiers, both Z_o and Z_i are resistive. Thus we represent Z_i by R_i and Z_o by R_o. The two port representation of the amplifier with its internal elements is shown in Fig. 3.86.

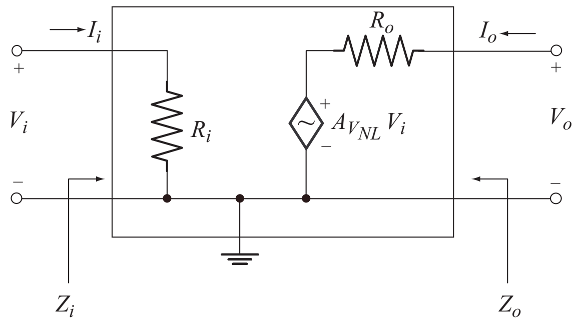


Fig. 3.86 Amplifier represented by two – port system

Now let us justify the placement of the parameters R_o and $A_{V_{NL}}$ in the output circuit. Applying KVL to the output circuit, we have

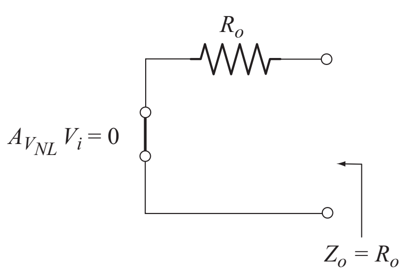
$$A_{V_{NL}} V_i + I_o R_o - V_o = 0.$$

Under no load condition (output open circuited), $I_o = 0$.

$$\therefore\ V_o = A_{V_{NL}} V_i = \text{open circuit output voltage.}$$

or $$A_{V_{NL}} = \frac{V_o}{V_i} = \text{no load or open circuit voltage gain.}$$

To find the output impedance, we set $V_i = 0$. As a result $A_{V_{NL}}\ V_i = 0$ which represents a short circuit. The resulting circuit is shown in Fig. 3.87.



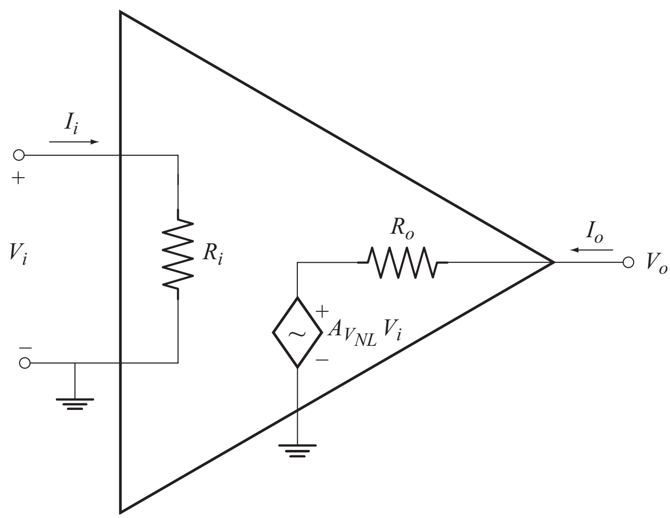


Fig. 3.87 justifying the placement of R_o **Fig. 3.88** Two port notation for op-amp

The circuit of Fig. 3.87 we find that $Z_o = R_o$ which justifies the placement of R_o.

A second format of two port representation of amplifiers which is popular with op-amps is shown in Fig. 3.88.

3.38 ANALYSIS OF AMPLIFIERS USING TWO – PORT SYSTEM APPROACH CONSIDERING THE EFFECTS OF R_S AND R_L

Figure 3.89 shows the two port representation of an amplifier driven by a source of internal resistance R_S and supplying the load R_L.

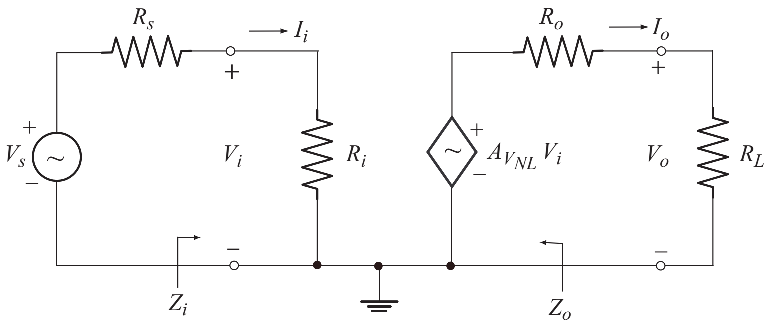


Fig. 3.89 Two - port system with R_S and R_L

Using voltage division rule in the input circuit we have

$$V_i = \frac{R_i}{R_S + R_i} V_S \tag{3.145}$$

or

$$\frac{V_i}{V_S} = \frac{R_i}{R_S + R_i} \tag{3.146}$$

Applying voltage division rule to the output circuit we get

$$V_o = \frac{R_L}{R_o + R_L} A_{V_{NL}} V_i \tag{3.147}$$

or

$$A_V = \frac{V_o}{V_i} = \frac{R_L}{R_o + R_L} A_{V_{NL}} \tag{3.148}$$

The total voltage gain is given by

$$A_{V_S} = \frac{V_o}{V_S}$$

$$= \frac{V_o}{V_i} \frac{V_i}{V_S} \tag{3.149}$$

Substituting Equations (3.146) and (3.148) into Equation (3.149) we have

$$A_{V_S} = \frac{R_i}{R_S + R_i} \cdot \frac{R_L}{R_o + R_L} A_{V_{NL}} \tag{3.150}$$

The current gain is given by

$$A_I = \frac{I_o}{I_i}$$

Using $I_o = -\dfrac{V_o}{R_L}$ and $I_i = \dfrac{V_i}{R_i}$ we get

$$A_I = \frac{(-V_o/R_L)}{(V_i/R_i)}$$

$$= -\frac{V_o}{V_i} \cdot \frac{R_i}{R_L}$$

$$A_I = -A_V \frac{R_i}{R_L} \tag{3.151}$$

The current gain taking R_S into account is given by

$$A_{I_S} = \frac{I_o}{I_i}$$

But $$I_i = \frac{V_S}{R_S + R_i}$$

$\therefore$ $$A_{I_S} = (-V_o / R_L) \div \frac{V_S}{R_S + R_i}$$

$$= -\frac{V_o}{V_S} \cdot \frac{R_S + R_i}{R_L}$$

$$A_{I_S} = -A_{V_S} \frac{R_S + R_i}{R_L} \tag{3.152}$$

It is important to note that $A_I = A_{I_S}$ when $R_S = 0$

3.39 ADVANTAGES OF TWO-PORT SYSTEM APPROACH

Now a days the amplifiers are available in the packaged form. Along with the packaged product, the manufacturer supplies the no load values of gain, input and output impedances. Thus the designer can quickly and efficiently find the results for loaded conditions without worrying about the internal components of the package.

Example 3.17

For the circuit of example 3.16

(a) Calculate the parameters of the two port system.

(b) Sketch the two port model

(c) Determine A_V, A_{V_S} and A_I

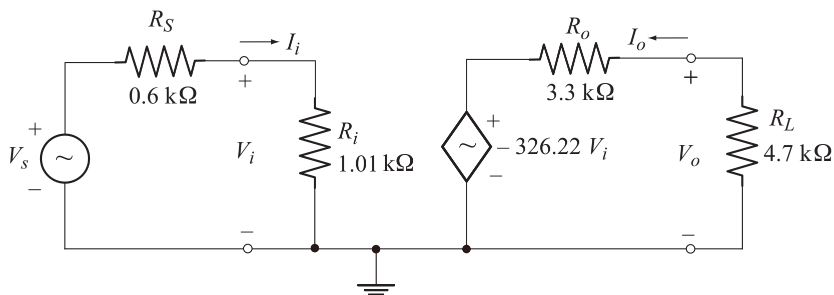
(d) Calculate A_V, A_{V_S} and A_I when $R_L = 2.7$ kΩ and $R_S = 0.3$ kΩ.

Solution

(a)

$$A_{V_{NL}} = -326.22$$
$$Z_i = R_i = 1.01\ \text{k}\Omega \qquad \text{[as calculated in example 3.16]}$$
$$Z_o = R_o = 3.3\ \text{k}\Omega$$

(b) The two port model is shown below.



(c)

$$A_V = \frac{R_L}{R_o + R_L} A_{V_{NL}}$$

$$= \frac{4.7\,\text{k}\Omega}{3.3\ \text{k}\Omega + 4.7\,\text{k}\Omega}(-326.22) = -191.65$$

$$A_{V_S} = A_V \frac{R_i}{R_S + R_i}$$

$$= (-191.65)\frac{1.01\text{k}\Omega}{0.6\text{k}\Omega + 1.01\ \text{k}\Omega} = -120.22$$

$$A_I = -A_V \frac{R_i}{R_L} = -(-191.65)\frac{1.01\text{k}\Omega}{4.7\,\text{k}\Omega} = 41.18$$

Note that these results agree with those obtained in example 3.16

(d) $R_L = 2.7\ \text{k}\Omega \qquad R_S = 0.3\ \text{k}\Omega$.

$$A_V = (-326.22)\frac{2.7\,\text{k}\Omega}{3.3\text{k}\Omega + 2.7\text{k}\Omega} = -146.79$$

$$A_{V_S} = (-146.79)\frac{1.01\ \text{k}\Omega}{0.3\ \text{k}\Omega + 1.01\ \text{k}\Omega} = -113.17$$

$$A_I = -(-146.79)\frac{1.01\ \text{k}\Omega}{2.7\ \text{k}\Omega} = 54.91$$

Note: Similarly we can analyse any BJT configuration using two-port system approach, with R_S and R_L.

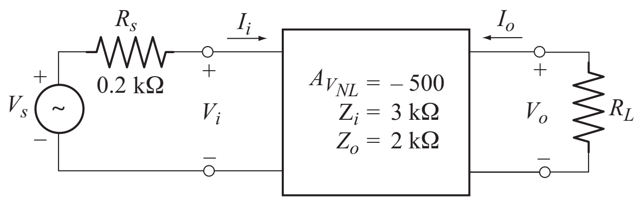
Example 3.18

For the packaged amplifier shown below:

(a) Calculate A_V for R_L = 1.2 kΩ and 5.6 kΩ and compare these values with $A_{V_{NL}}$.

(b) Calculate A_{V_S} with R_L = 1.2kΩ.

(c) Determine A_I with R_L = 5.6 kΩ.



Solution:

(a)
$$A_V = \frac{R_L}{R_o + R_L} A_{V_{NL}}$$

with R_L = 1.2 kΩ
$$A_V = \frac{1.2\,\text{k}\Omega}{2\,\text{k}\Omega + 1.2\,\text{k}\Omega}(-500)$$
$$= -187.5$$

with R_L = 5.6 kΩ
$$A_V = \frac{5.6\,\text{k}\Omega}{2\,\text{k}\Omega + 5.6\,\text{k}\Omega}(-500)$$
$$= -368.42$$

Note that as R_L decreases, A_V also decreases. It is important to note that $A_V \to A_{V_{NL}}$ for $R_L \gg R_o$.

(b)
$$A_{V_S} = \frac{R_i}{R_S + R_i} A_V$$

with R_L = 1.2 kΩ, A_V = – 187.5
$$A_{V_S} = \frac{3\,\text{k}\Omega}{0.2\,\text{k}\Omega + 3\,\text{k}\Omega}(-187.5)$$
$$= -175.78.$$

Note that $A_{V_S} \to A_V$ for $R_i \gg R_S$.

(c)
$$A_I = -\frac{A_V Z_i}{R_L}$$

with R_L = 5.6 kΩ, A_V = – 368.42
$$A_I = -\frac{(-368.42)(3\,\text{k}\Omega)}{5.6\,\text{k}\Omega} = 197.36.$$

Exercise Problems

3.1 A CE fixed bias configuration has $R_C = 2\ \text{k}\Omega$, $R_B = 330\ \text{k}\Omega$, $\beta = 100$, $r_o = 50\ \text{k}\Omega$ and $V_{CC} = 15$ V. Calculate Z_i, Z_o, A_V and A_I.

3.2 A CE emitter-bias configuration with unbypassed R_E has $R_C = 2.7\ \text{k}\Omega$, $R_B = 390\ \text{k}\Omega$, $R_E = 1.2\ \text{k}\Omega$, $\beta = 150$, $r_o = 25\ \text{k}\Omega$ and $V_{CC} = 16$ V. Calculate Z_i, Z_o, A_V and A_I.

3.3 The following component values are available for an emitter follower.

$R_1 = 62\ \text{k}\Omega$, $R_2 = 10\ \text{k}\Omega$, $R_E = 1.5\ \text{k}\Omega$, $\beta = 200$, $r_o = 50\ \text{k}\Omega$ and $V_{CC} = 18$ V. Calculate Z_i, Z_o, A_V and A_I.

3.4 A CB amplifier has $R_C = 3.9\ \text{k}\Omega$, $R_E = 3.3\ \text{k}\Omega$, $V_{CC} = 9$ V, $V_{EE} = 4$ V, $\beta = 100$. Calculate Z_i, Z_o, A_V and A_I.

3.5 A CE collector dc feedback configuration has $V_{CC} = 14$ V, $R_C = 3.3\ \text{k}\Omega$, $R_{F_1} = 120\ \text{k}\Omega$, $R_{F_2} = 68\ \text{k}\Omega$, $\beta = 140$, and $r_o = 50\ \text{k}\Omega$. Calculate Z_i, Z_o, A_V and A_I.

Chapter 4

Transistor Frequency Response

◆ INTRODUCTION

The frequency response of an amplifier is the plot of the magnitude of voltage gain as a function of frequency. In transistor amplifier the low frequency response is governed by the coupling and bypass capacitors. The high frequency response is affected by the transistor parasitic capacitances and the stray wiring capacitances. The mid frequency response is unaffected by these capacitances.

This chapter discusses the effect of coupling and bypass capacitors on low frequency response. The effect of transistor parasitic capacitances and wiring capacitances has also been considered.

◆ 4.1 GENERAL FREQUENCY CONSIDERATIONS

The response of a single stage or multistage amplifier depends on the frequency of the applied signal. The coupling and bypass capacitors affect the low frequency response since the reactance of these capacitors decreases with increase in frequency. The internal capacitances of the active devices (BJT or FET) and the stray wiring capacitances will limit the high frequency response of the system. An increase in the number of stages of a cascaded system will also limit the low and high frequency responses.

4.1.1 Frequency Response of R-C Coupled Amplifier

The frequency response of an amplifier is the plot of the magnitude of voltage gain as a function of frequency. Figure 4.1 shows the frequency response of R-C coupled amplifier.

The scale on horizontal axis is a logarithmic scale to facilitate the plot extending from the low to the high frequency regions.

The frequency range is divided into three regions

- Low frequency region
- Mid frequency region
- High frequency region.

The drop in the gain at low frequencies is due to the coupling capacitors (C_C and C_S) and bypass capacitors (C_E).

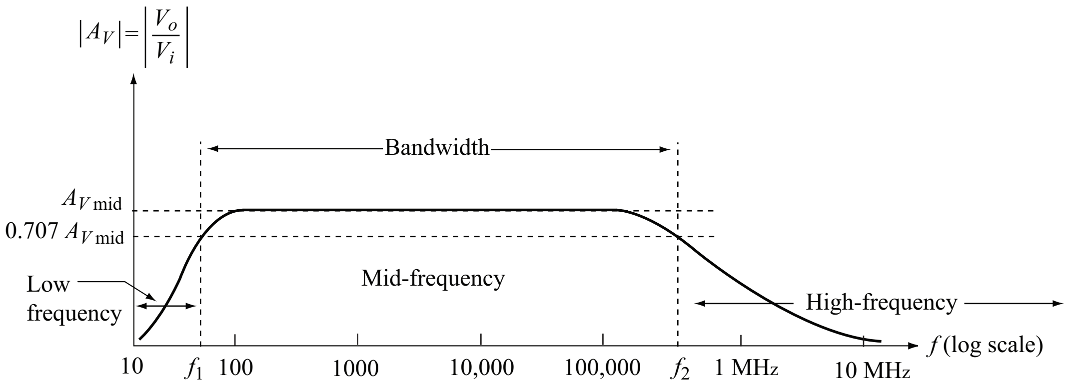


Fig. 4.1 Frequency response of R-C coupled amplifier

At high frequencies the drop in gain is due to the internal device capacitances and the stray wiring capacitances.

In the mid frequency range the gain is almost independent of the frequency. This is due to the fact that at mid frequencies the coupling and bypass capacitors act as short circuits and the device and stray wiring capacitances act as open circuits due to their low capacitance. The mid band gain is denoted by $A_{V\,mid}$.

4.1.2 Frequency Response of Transformer Coupled Amplifier

Figure 4.2 shows the frequency response of transformer coupled amplifier.

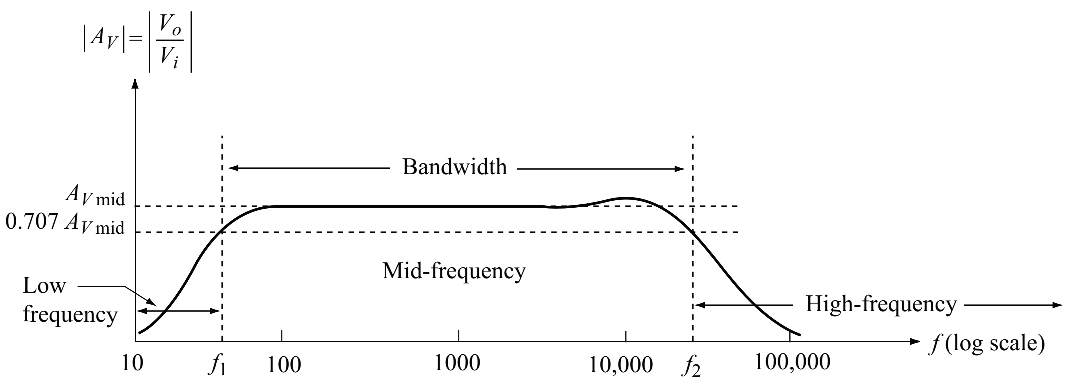


Fig 4.2 Frequency response of transformer coupled amplifier

The magnetising inductive reactance of the transformer winding is $X_L = 2\,\pi f L$.

At low frequencies the gain drops due to the small value of X_L. At $f = 0$ (DC) there is no change in flux in the core. As a result the secondary induced voltage or output voltage is zero and hence the gain.

At high frequencies the gain drops due to the stray capacitance between the turns of primary and secondary windings.

4.1.3 Frequency Response of Direct Coupled Amplifier

Figure 4.3 shows the frequency response of direct coupled amplifier.

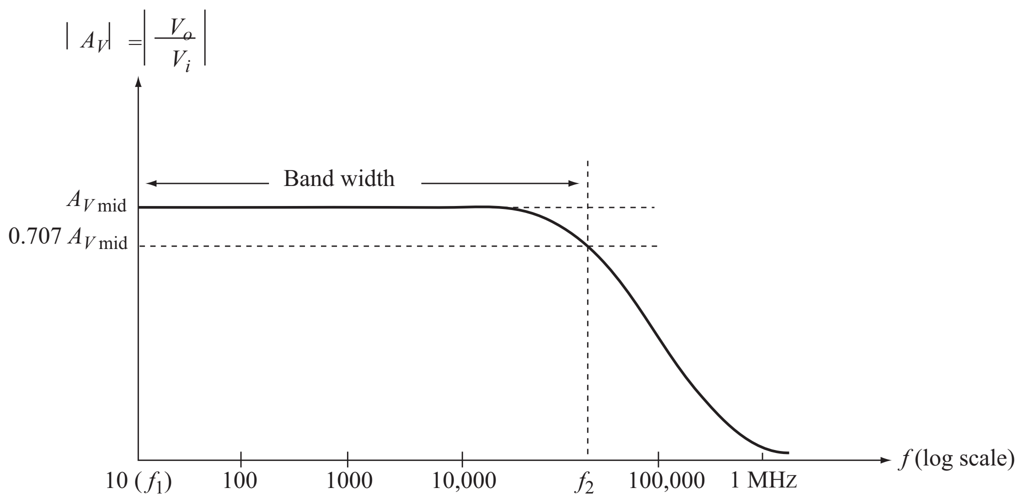


Fig. 4.3 Frequency response of direct coupled amplifier

Direct coupled amplifier do not use coupling and bypass capacitors. As a result there is no drop in gain at low frequencies. The frequency response curve is flat upto the upper cutoff frequency.

Gain drops at high frequencies due to the device internal capacitances and the stray wiring capacitances.

4.1.4 Half Power Frequencies and Band Width

The frequencies f_1 and f_2 at which the gain is 0.707 $A_{V\,\text{mid}}$ are called cut-off frequencies or corner frequencies or break frequencies. f_1 is called the lower cut-off frequency and f_2 the upper cut-off frequency.

The band width or the pass band of the amplifier is given by

Band width, $$BW = f_2 - f_1 \tag{4.1}$$

The output voltage in the mid band is

$$|V_o| = |A_{V\,\text{mid}}|\ |V_i|$$

Output power in the mid band is

$$P_{o\,(\text{mid})} = \frac{|V_o|^2}{R_o}$$

$$= \frac{|A_{V\,\text{mid}}|^2\,|V_i|^2}{R_o} \tag{4.2}$$

The output voltage at cut-off frequencies is

$$|V_o| = |0.707\,A_{V\,\text{mid}}|\,|V_i|$$

and the output power at cut-off frequencies is

$$P_{o\,(\text{cut-off})} = \frac{|0.707 A_{V\,\text{mid}}|^2 |V_i|^2}{R_o}$$

$$= 0.5\,\frac{|A_{V\,\text{mid}}|^2 |V_i|^2}{R_o}$$

or $$P_{o\,(\text{cut-off})} = 0.5\,P_{o\,(\text{mid})} \tag{4.3}$$

Note that the output power at cut-off frequencies is half the mid band power output. For this reason f_1 and f_2 are also called the half power frequencies. More specifically f_1 is called the lower half-power frequency and f_2 the upper half power frequency.

4.1.5 Normalised Gain versus Frequency Plot

The normalized gain is obtained by dividing the gain A_V at each frequency by the mid band gain $A_{V\,\text{mid}}$. Therefore

$$\text{Normalised gain} = \frac{A_V}{A_{V\,\text{mid}}} \tag{4.4}$$

Figure 4.4 shows the normalized gain versus frequency plot for an RC-coupled amplifier. Note that:

- The normalised mid band gain is $\dfrac{A_{V\,\text{mid}}}{A_{V\,\text{mid}}} = 1$ and
- The normalised gain at cut-off frequencies is $\dfrac{0.707 A_{V\,\text{mid}}}{A_{V\,\text{mid}}} = 0.707.$

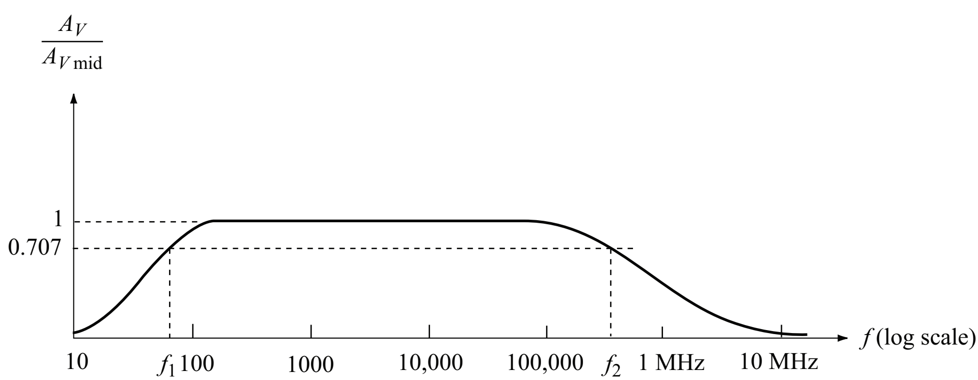


Fig. 4.4 Normalised gain versus frequency plot

In communication applications such as audio and video it is more useful to plot the normalised decibel voltage gain versus frequency rather than the normalised voltage gain versus frequency.

Normalised decibel voltage gain is

$$\left.\frac{A_V}{A_{V\,\mathrm{mid}}}\right|_{dB} = 20\log_{10}\left[\frac{A_V}{A_{V\,\mathrm{mid}}}\right] \tag{4.5}$$

Normalised decibel voltage gain in mid band is

$$20\log_{10}\left[\frac{A_{V\,\mathrm{mid}}}{A_{V\,\mathrm{mid}}}\right] = 0$$

Normalised decibel voltage gain at cut-off frequencies is

$$20\log_{10}\left[\frac{0.707\,A_{V\,\mathrm{mid}}}{A_{V\,\mathrm{mid}}}\right] = -3\text{ dB}.$$

Note that normalised decibel voltage gain at cut-off frequencies is 3dB less than the normalised decibel midband voltage gain. For this reason the frequencies f_1 and f_2 are also called the 3 dB frequencies. More specifically f_1 is called the lower 3 dB frequency and f_2 the upper 3 dB frequency.

Figure 4.5 shows the plot of normalised decibel voltage gain versus frequency for an RC-coupled amplifier.

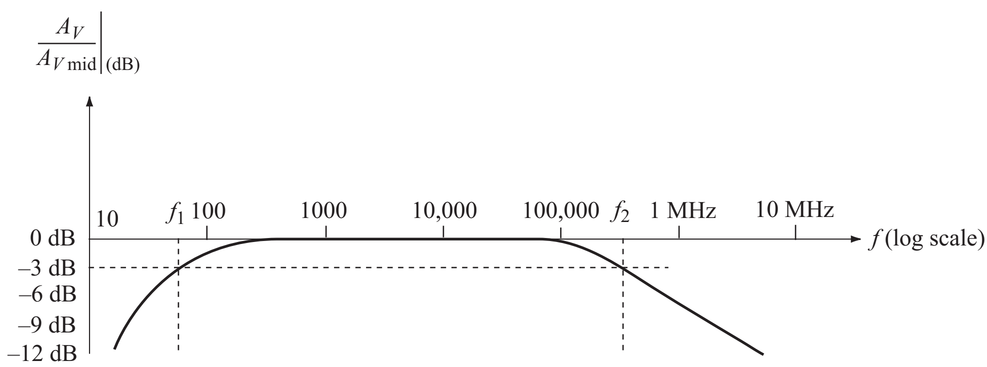


Fig. 4.5 Plot of normalised decibel voltage gain versus frequency

4.1.6 Phase Angle Plot

A single stage RC coupled amplifier introduces a 180° phase shift between input and output signals in the mid band region. At low frequencies the output voltage V_o lags V_i by an additional angle θ_1. Therefore the total phase shift between V_o and V_i is more than 180°. At high frequencies V_o leads V_i by an additional angle θ_2. As a result the total phase shift drops below 180°. Figure 4.6 shows the phase plot for a single stage RC-coupled amplifier.

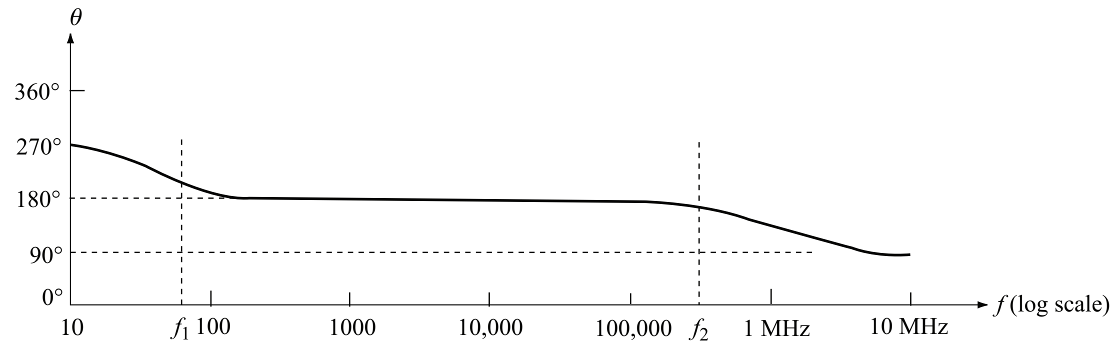


Fig. 4.6 Phase plot of single stage RC-coupled amplifier

◆ 4.2 LOW FREQUENCY ANALYSIS

In low frequency region, we have seen that the amplifier gain increases with frequency. Hence it can be modelled as a high-pass RC circuit as shown in Fig. 4.7.

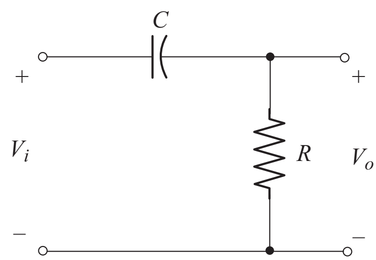


Fig. 4.7 Amplifier modelled as high-pass RC circuit

The capacitor C represents the combined effect of coupling and bypass capacitors and the resistance R represents the combined effect of resistive elements of the amplifier network.

The capacitive reactance is given by

$$X_C = \frac{1}{2\pi f C} \tag{4.6}$$

At $f = 0$, $X_C = \infty\ \Omega$

i.e. at low frequencies the capacitor acts as an open circuit as shown in Fig. 4.8.

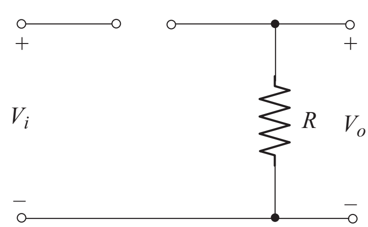


Fig. 4.8 Capacitor acts as open circuit at f = 0

From the circuit of Fig. 4.8 we find that, $V_o = 0$.
At high frequencies, $X_C \approx 0\ \Omega$
i.e., at high frequencies the capacitor acts as a short circuit as shown in Fig. 4.9.

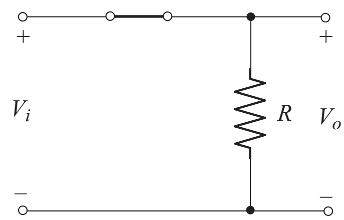


Fig. 4.9 Capacitor acts as short circuit at high frequencies

Form the circuit of Fig. 4.9, we find that, $V_o \approx V_i$. Note that as the input signal frequency increases from zero to the mid band value, the output voltage rises from zero to V_i and hence the gain from zero to one. Let us verify this fact using mathematical analysis.

Mathematical Analysis

Using voltage division rule in the circuit of Fig. 4.7, we have

$$V_o = \frac{V_i\, R}{R - jX_C}$$

Voltage gain is given by

$$A_V = \frac{V_o}{V_i} = \frac{R}{R - jX_C}$$

$$A_V = \frac{1}{1 - j\left[\dfrac{X_C}{R}\right]} \tag{4.7}$$

The magnitude of voltage gain is

$$|A_V| = \frac{1}{\sqrt{1 + \left[\dfrac{X_C}{R}\right]^2}} \tag{4.8}$$

Now let us find the gain at different frequencies:

- At $f = 0$,
$$X_C = \frac{1}{2\pi f C} = \infty\ \Omega$$
$$\therefore \quad |A_V| = 0$$

- At high frequencies
$f \to \infty$ and therefore $X_C \to 0$

as a result, $|A_V| \to 1 = |A_V|_{\text{mid}}$

Now $|A_V|_{\text{mid (dB)}} = 20 \log_{10}(1) = 0 \text{ dB}.$

- When the capacitive reactance equals the resistance

$$\text{ie} \qquad X_C = R \tag{4.9}$$

$$|A_V| = \frac{1}{\sqrt{2}} \Rightarrow \frac{V_o}{V_i} = \frac{1}{\sqrt{2}} \quad \text{or} \quad V_o = 0.707\, V_i$$

The corresponding decibel gain is

$$20 \log_{10} \frac{1}{\sqrt{2}} = -3 \text{ dB}$$

Note that this condition must give the cut-off frequency
From Equation (4.9)

$$\frac{1}{2\pi f C} = R$$

$$f = \frac{1}{2\pi RC}$$

The frequency given by the above equation is the lower cut-off frequency or the lower 3dB cut-off frequency denoted by f_1.

$$\therefore \qquad f_1 = \frac{1}{2\pi RC} \tag{4.10}$$

$$\frac{X_C}{R} = \frac{1}{2\pi f CR} = \left[\frac{1}{2\pi RC}\right]\left[\frac{1}{f}\right] = \frac{f_1}{f}$$

Using this relation in Equations (4.7) and (4.8) we have

$$A_V = \frac{1}{1 - j\left[\dfrac{f_1}{f}\right]} \tag{4.11}$$

$$|A_V| = \frac{1}{\sqrt{1 + \left[\dfrac{f_1}{f}\right]^2}} \tag{4.12}$$

From Equation (4.11) the phase angle of A_V is

$$\theta_1 = \tan^{-1}\left[\frac{f_1}{f}\right] \tag{4.13}$$

Since θ_1 is positive, V_o leads V_i by an angle θ_1.

In magnitude and phase form, Equation (4.11) can be written as

$$A_V = |A_V| \angle \theta_1$$

$$A_V = \frac{1}{\sqrt{1+\left[\frac{f_1}{f}\right]^2}} \angle \tan^{-1}\left[\frac{f_1}{f}\right] \tag{4.14}$$

Figure 4.10 shows the plot of $|A_V|$ versus frequency.

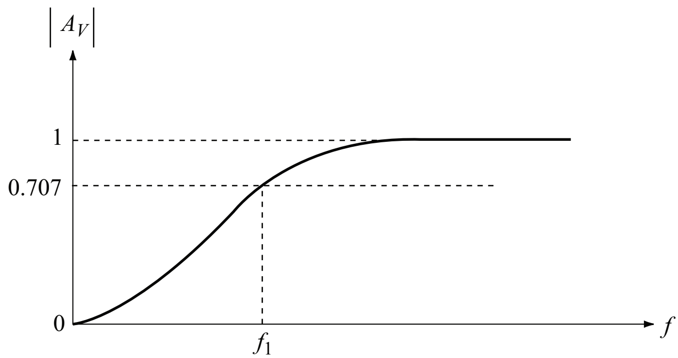


Fig. 4.10 Low frequency response of high pass-RC circuit

Bode Plot of Low Frequency Response

From Equation (4.12) we have

$$|A_V| = \frac{1}{\sqrt{1+\left[\frac{f_1}{f}\right]^2}}$$

Voltage gain in dB is

$$|A_V|_{\text{dB}} = 20\log_{10}\left[\frac{1}{\sqrt{1+\left[\frac{f_1}{f}\right]^2}}\right]$$

$$= -20\log_{10}\sqrt{1+\left[\frac{f_1}{f}\right]^2} \tag{4.15}$$

Now let us construct the plot of $|A_V|_{\text{dB}}$ versus frequency using the straight line segments by considering the following frequency ranges.

1. **For frequencies,** $f \ll f_1$ **or** $\dfrac{f_1}{f} \gg 1$

Equation (4.15) can be approximated by

$$|A_V|_{\text{dB}} \approx -20 \log_{10} \sqrt{\left[\frac{f_1}{f}\right]^2}$$

or
$$|A_V|_{\text{dB}} = -20 \log_{10} \left[\frac{f_1}{f}\right] \qquad (4.16)$$

$|A_V|_{\text{dB}}$ is calculated at different values of $\dfrac{f_1}{f}$ and tabulated in Table 4.1.

Table 4.1 $|A_V|_{\text{dB}}$ at different frequencies

f	$\dfrac{f_1}{f}$	$|A_V|_{\text{dB}} = -20 \log_{10}\left[\dfrac{f_1}{f}\right]$
$\dfrac{f_1}{10}$	10	– 20 dB
$\dfrac{f_1}{4}$	4	– 12 dB
$\dfrac{f_1}{2}$	2	– 6 dB
f_1	1	0 dB

From the results given in Table 4.1, we can draw the following interesting conclusions.

- A change in frequency by a factor of two is equal to one octave. When the frequency changes from $\dfrac{f_1}{4}$ to $\dfrac{f_1}{2}$ or $\dfrac{f_1}{2}$ to f_1 (one octave), the gain increases by 6 dB.
- A change in frequency by a factor of ten, is equal to one decade. When the frequency changes from $\dfrac{f_1}{10}$ to f_1 (one decade), the gain increases by 20 dB.
- If we plot $|A_V|_{\text{dB}}$ against log scale in the frequency range $\dfrac{f_1}{10} < f < f_1$, we get a straight line with slope 6 dB/octave or 20 dB/decade, as shown in Fig. 4.11.

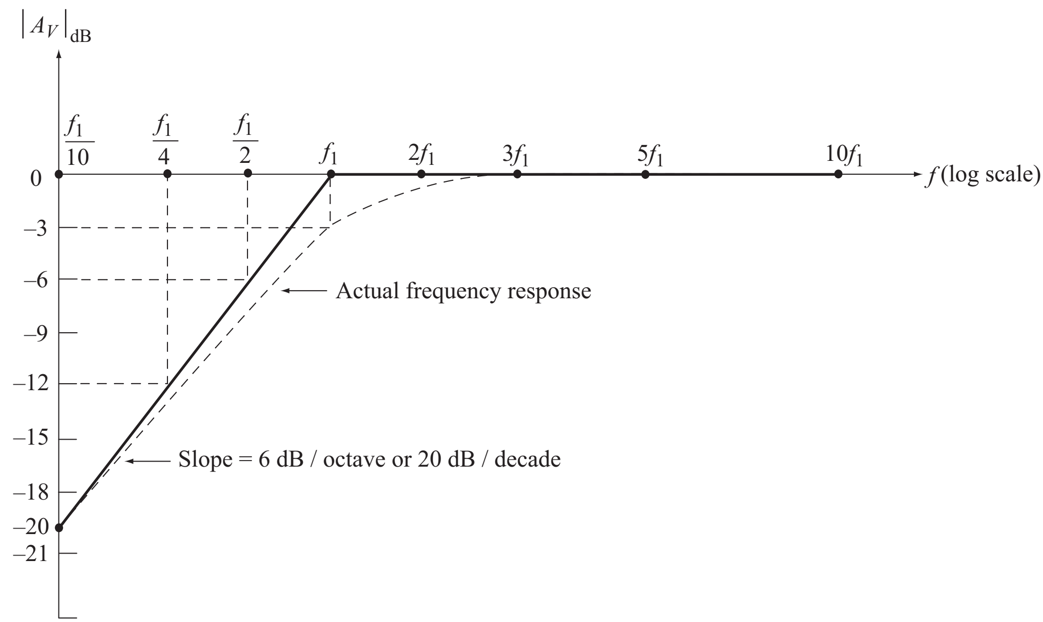


Fig. 4.11 Bode plot for low frequency region

2. **For frequencies,** $\quad f \gg f_1 \;$ **or** $\; \dfrac{f_1}{f} \ll 1$

Equation (4.15) can be approximated by

$$|A_V|_{dB} \approx -20 \log_{10} 1 = 0 \text{ dB}$$

The plot of $|A_V|_{dB}$ against log scale for the frequency range $f \gg f_1$, is a straight line on the frequency axis as shown in Fig 4.15. Note that the slope of this line is zero since the gain is constant at 0 dB.

The plot shown in Fig 4.11 is made up of two straight line segments called asymptotes with a break point at f_1. Hence f_1 is called the break frequency or the corner frequency. This piece wise linear plot is also called the Bode magnitude plot or simply Bode plot.

The actual frequency response is also indicated in Fig. 4.11. Note that

From Bode plot, at $f = f_1$, $|A_V|_{dB} = 0$.

From actual plot, at $f = f_1$, $|A_V|_{dB} = -3$.

We find that at $f = f_1$, the gain read from the Bode plot differs from the actual gain by 3 dB.

Phase Plot

At low frequencies, V_o leads V_i by an angle θ_1 given in Equation (4.13) by

$$\theta_1 = \tan^{-1}\left[\frac{f_1}{f}\right] \tag{4.17}$$

The value of θ_1 is calculated at different values of $\frac{f_1}{f}$ and tabulated in Table 4.2.

Table 4.2 Phase angle between V_o and V_i

f	$\frac{f_1}{f}$	$\theta_1 = \tan^{-1}\left[\frac{f_1}{f}\right]$	*Total phase shift* $\theta = 180 + \theta_1$
0	∞	90^o	270^o
$\frac{f_1}{100}$	100	89.4^o	269.4^o
f_1	1	45^o	225^o
$100 f_1$	0.01	0.572^o	180.572^o
∞	0	0^o	180^o

The total phase shift θ between V_o and V_i is the sum of the phase shift of RC network and the inherent phase shift (180°) introduced by the amplifier.

The following observations can be made from the results given in Table 4.2.

When the input signal frequency increases from zero to the mid band value ($f \gg f_1$).

- The phase shift θ_1 due to RC network decreases from 90° to 0°. The plot of θ_1 versus frequency is shown in Fig. 4.12.
- The total phase θ, decreases from 270° to 180°.

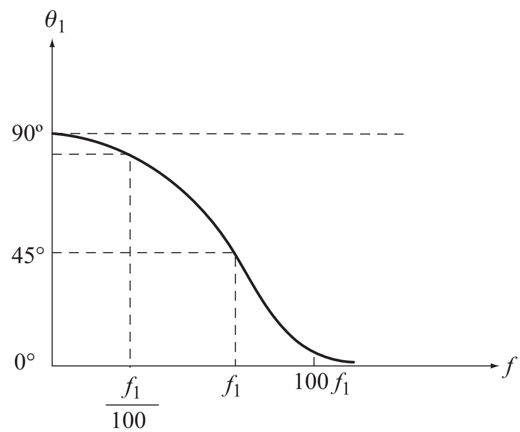


Fig. 4.12 Phase response of RC network of Fig 4.7

Example 4.1

For the circuit shown below

(a) Determine the mathematical expression for $\left|\frac{V_o}{V_i}\right|$

(b) Calculate the break frequency

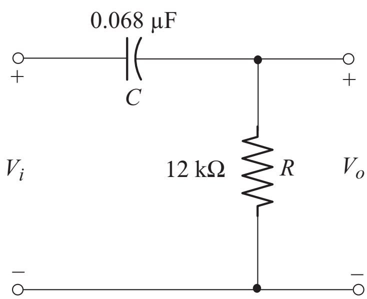
(c) Calculate $\left|\frac{V_o}{V_i}\right|$ at 10 Hz, 100 Hz, 1 kHz, 2 kHz, 5 kHz and 10 kHz

(d) Sketch the frequency response of $\left|\frac{V_o}{V_i}\right|$

(e) Construct the Bode magnitude plot

(f) Sketch the actual frequency response

(g) Compare the results obtained in part (d) and part (e).



Solution

(a) The expression for $\left|\frac{V_o}{V_i}\right|$ is given in Equation (4.12) which is derived under mathematical analysis in section 4.2.

$$|A_V| = \left|\frac{V_o}{V_i}\right| = \left[\frac{1}{\sqrt{1+\left[\frac{f_1}{f}\right]^2}}\right]$$

(b) The break frequency is

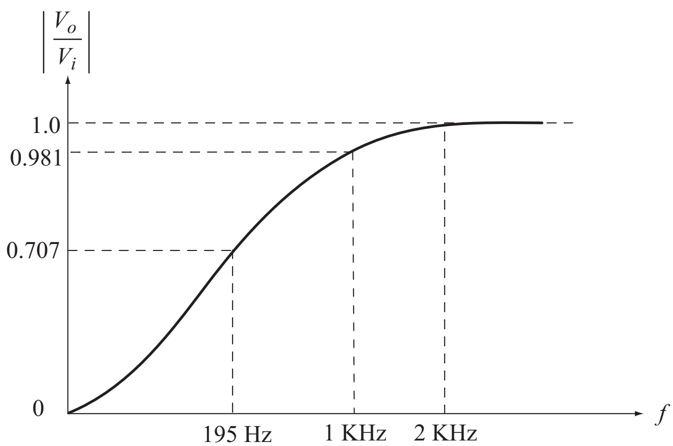
$$f_1 = \frac{1}{2\pi RC}$$

$$= \frac{1}{(2)(\pi)(12\ \text{k}\Omega)(0.068\ \mu\text{F})} = 195\ \text{Hz}.$$

(c) Table A

f	$\dfrac{f_1}{f}$	$\lvert A_V \rvert = \left\lvert \dfrac{V_o}{V_i} \right\rvert$	$\lvert A_V \rvert_{dB} = 20 \log_{10} \lvert A_V \rvert$
10 Hz	19.5	0.0512	– 25.8
19.5 Hz $\left[= \dfrac{f_1}{10}\right]$	10	0.0995	– 20.04
100 Hz	1.95	0.456	– 6.82
195 Hz $[= f_1]$	1	0.707	– 3
1 KHz	0.195	0.981	– 0.166
2 KHz	0.0975	1	0
5 KHz	0.039	1	0
10 KHz	0.0195	1	0

(d) Frequency response of $\left|\dfrac{V_o}{V_i}\right|$



(e) Bode plot

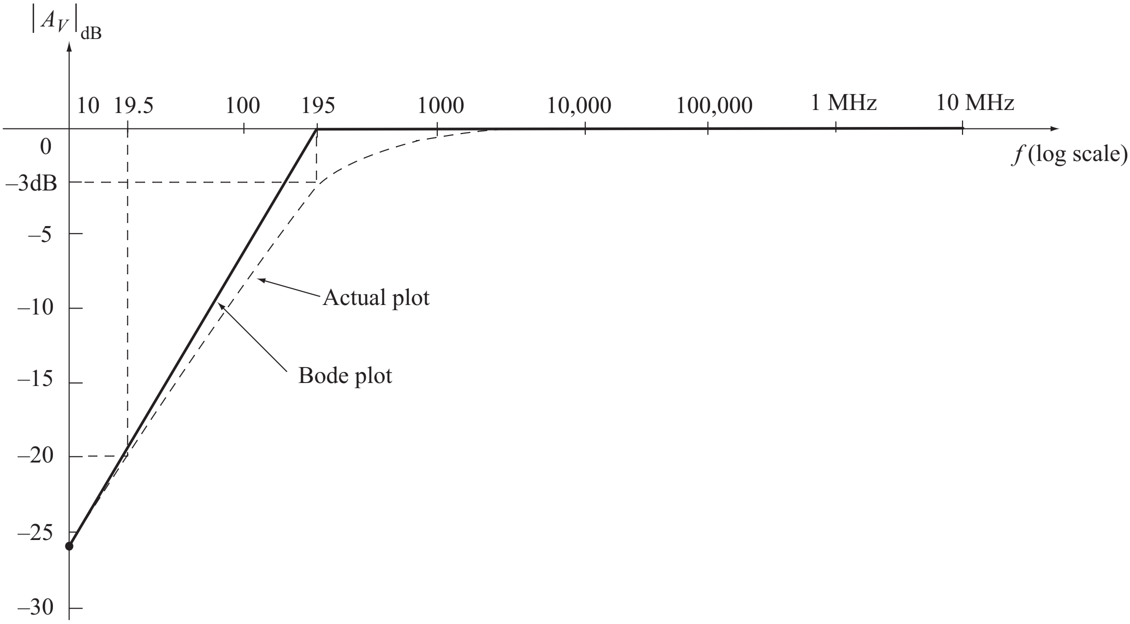
It consists of two straight line segments.

at $f = \dfrac{f_1}{10}$ $\qquad \lvert A_V \rvert_{dB} = -20 \log_{10} \left[\dfrac{f_1}{f}\right] = -20$

at $f = f_1$ $\qquad \lvert A_V \rvert_{dB} = -20 \log_{10} [1] = 0$

The first line is obtained by joining these two points.
This line has a slope of 20 dB/decade or 6 dB/octave.
The second line is drawn on frequency axis starting from $f = f_1$ at it has slope of 0 dB/decade.

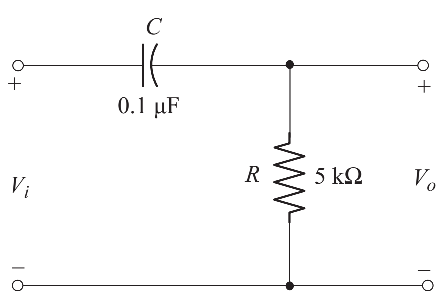
(f) The actual plot is sketched using the values given in the 4th column of Table A, along with Bode plot in the following figure.



Example 4.2

For the circuit shown below

(a) Determine the mathematical expression for the phase angle θ_1 between V_o and V_i.
(b) Calculate the break frequency.
(c) Calculate phase angle θ_1 at f = 100 Hz, 1 KHz, 2 KHz, 5 KHz and 10 KHz.
(d) Sketch the phase angle versus frequency plot using the results obtained in part (c).



Solution

(a) The expression for θ_1 is given in Equation (4.13) which is derived under mathematical analysis in section 4.2.

$$\theta_1 = \tan^{-1}\left[\frac{f_1}{f}\right]$$

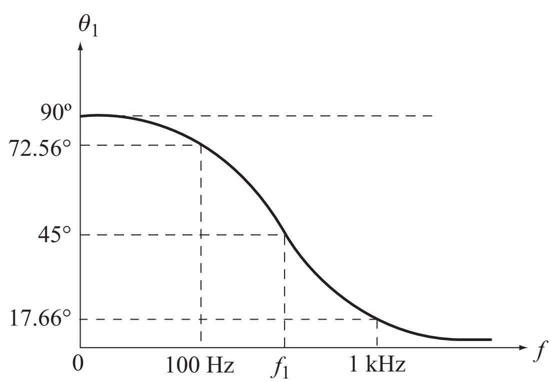
(b) The break frequency is

$$f_1 = \frac{1}{2\pi RC} = \frac{1}{2\pi(5\ \text{k}\Omega)(0.1\ \mu\text{F})} = 318.3\ \text{Hz}$$

(c) Calculation of θ_1 at different frequencies

f	$\frac{f_1}{f}$	$\theta_1 = \tan^{-1}\left[\frac{f_1}{f}\right]$
0	∞	90°
100 Hz	3.183	72.56°
318.3 Hz [$= f_1$]	1	45°
1 KHz	0.3183	17.66°
2 KHz	0.15915	9.043°
5 KHz	0.06366	3.643°
10 KHz	0.03183	1.823°

(d) The phase angle plot is shown below



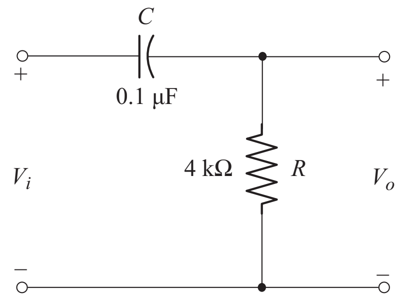
Example 4.3

For the circuit shown below

(a) Calculate the lower 3 dB frequency.

(b) Find the magnitude and phase angle of A_V at 1000 Hz.

(c) Calculate the output voltage if the input voltage is
$v_i = 10 \sin [12566\, t]$ V



Solution

(a) Lower 3 dB frequency is

$$f_1 = \frac{1}{2\pi RC} = \frac{1}{2\pi (4\text{ k}\Omega)(0.1\ \mu\text{F})} = 397.88 \text{ Hz}$$

(b) Magnitude of A_V is

$$|A_V| = \frac{1}{\sqrt{1+\left[\frac{f_1}{f}\right]^2}}$$

Phase angle of A_V is

$$\theta_1 = \tan^{-1}\left[\frac{f_1}{f}\right]$$

At $f = 1000$Hz

$$|A_V| = \frac{1}{\sqrt{1+\left[\frac{397.88}{1000}\right]^2}} = 0.929$$

$$\theta_1 = \tan^{-1}\left[\frac{397.88}{1000}\right] = 21.69°$$

(c)

$$v_i = 10 \sin [12566\, t] = V_m \sin \omega t$$

$$V_m = 10 \text{ V}$$

$$\omega = 12566 \text{ rad/s}$$

$$f = \frac{\omega}{2\pi} = \frac{12566}{2\pi} = 2000 \text{ Hz}$$

$$|A_V| = \frac{1}{\sqrt{1+\left[\frac{397.88}{2000}\right]^2}} = 0.98$$

$$\theta_1 = \tan^{-1}\left[\frac{397.88}{2000}\right] = 11.25°$$

v_o leads v_i by an angle θ_1

$$\therefore \quad v_o = V_m \cdot |A_V| \cdot \sin[\omega t + \theta_1]$$

$$= [10][0.98] \sin[12566\, t + 11.25°]$$

$$= 9.8 \sin[12566\, t + 11.25°] \text{ V}$$

Example 4.4

It is desired that the voltage gain of an RC-coupled amplifier at 60 Hz should not decrease by more than 10% from its mid band value. Calculate

(a) the lower 3dB frequency
(b) the required C if $R = 2000\ \Omega$

Solution

(a) Note that 60 Hz lies in the low frequency region

$$\therefore \quad |A_V| = \frac{1}{\sqrt{1+\left[\frac{f_1}{f}\right]^2}} \qquad \text{(A)}$$

Given at $\quad f = 60$ Hz, $\quad |A_V| = 0.9$

Using this condition in Equation (A), we get

$$0.9 = \frac{1}{\sqrt{1+\left[\frac{f_1}{60}\right]^2}}$$

Solving we get, $f_1 = 29$ Hz

(b)

$$f_1 = \frac{1}{2\pi RC}$$

$$C = \frac{1}{2\pi f_1 R} = \frac{1}{2\pi(29\,\text{Hz})(2000\,\Omega)}$$

$$= 2.74\ \mu\text{F}$$

◆ 4.3 LOW FREQUENCY RESPONSE OF BJT AMPLIFIER

Figure 4.13 shows the circuit of single stage BJT amplifier. The coupling capacitors C_S and C_C and the bypass capacitor C_E determine the low frequency response. Now let us study the influence of each of these on low frequency response.